What is Data Sciences (*Application: Machine Learning*)

Broadly, machine learning is the application of **statistical, mathematical, and numerical** techniques to **derive some form of knowledge** from data. This ‘knowledge’ may afford us some sort of summarization, visualization, grouping, or even predictive power over data sets.

With all that said, it’s important to emphasize the limitations of machine learning.

It is not nor will it ever be a replacement for critical thought and methodical, procedural work in **data science**. Indeed, machine learning can be reasonably characterized a loose collection of disciplines and tools.

Regression

A major component of machine learning **(Data Science)**, the one that most people associate with ML, is dedicated to making predictions about a target given some inputs, such as predicting how much money an individual will earn in their lifetime given their demographic information.

In this part, one is going to focus on the case where **its prediction is a continuous, real number**.

When the target is a real number, one call this prediction procedure regression.

In order to understand regression, let's start in a more natural place:
What types of problems are we trying to solve?
What exactly does it mean to make a prediction that is a continuous, real number?
To build some intuition, here are a few examples of problems that regression could be used for:

1. Predicting a person's height given the height of their parents.
2. Predicting the amount of time someone will take to pay back a loan given their credit history.
3. Predicting what time a package will arrive given current weather and traffic conditions.

Given some inputs, one needs to produce a prediction for a continuous output.
That is exactly the purpose of regression.
Notice that regression isn't any one technique in particular. It's just a class of methods that helps us achieve our overall goal of predicting a continuous output.

Solution Options

<u>List of regression techniques</u>

K-Nearest-Neighbors (KNN)

K-Nearest-Neighbors is an extremely intuitive, non-parametric technique for regression or classification.

It works as follows in the regression case:

1. Identify the K points in our data set that are closest to the new data point. 'Closest' is some measure of distance, usually **Euclidean**.
2. **Average** the value of interest for those K data points.
3. Return **that averaged value of interest**: it is the prediction for our new data point.

Neural Networks

A neural network works by scaling and combining input variables many times. Furthermore, at each step of scaling and combining inputs, we typically apply some form of nonlinearity over our data values.

Random Forests

Random forests are what's known as an ensemble method. This means we average the results of many smaller regressors known as decision trees to produce a prediction. These decision trees individually operate by partitioning our original data set with respect to the value of predictive interest using a subset of the features present in that data set.

Gradient Boosted Trees

Gradient boosted trees are another technique built on decision trees. Assume we start with a set of decision trees that together achieve a certain level of performance.

Turning to Linear Regression

Linear Regression

Linear regression is a statistical method that is used in various machine learning models to predict the value of unknown data using other related data values. Linear regression is used to study the relationship **between a dependent variable and an independent variable.**

Linear Regression Equation

$$Y = a + bx$$

$$a = \frac{[(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

$$b = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

Definition 2.3.1 (Linear Regression): Suppose we have an input $\mathbf{x} \in \mathbb{R}^D$ and a continuous target $y \in \mathbb{R}$. Linear regression determines weights $w_i \in \mathbb{R}$ that combine the values of x_i to produce y:

$$y = w_0 + w_1 x_1 + ... + w_D x_D \tag{2.1}$$

⋆ Notice w_0 in the expression above, which doesn't have a corresponding x_0 value. This is known as the *bias* term. If you consider the definition of a line $y = mx + b$, the bias term is corresponds to the intercept b. It accounts for data that has a non-zero mean.

Let's illustrate how linear regression works using an example, considering the case of 10 year old Sam. She is curious about how tall she will be when she grows up. She has a data set of parents' heights and the final heights of their children. The inputs x are

$$x_1 = \text{height of mother (cm)}$$
$$x_2 = \text{height of father (cm)}$$

Using linear regression, she determines the weights $\mathbf{w}$ to be:

$$\mathbf{w} = [34, 0.39, 0.33]$$

Sam's mother is 165 cm tall and her father is 185 cm tall. Using the results of the linear regression solution, Sam solves for her expected height:

$$\text{Sam's height} = 34 + 0.39(165) + 0.33(185) = \mathbf{159.4\ cm}$$

Linear Regression Equation

Linear regression line equation is written in the form: **y = a + bx**

where,

→ x is Independent Variable, Plotted along X-axis

→ y is Dependent Variable, Plotted along Y-axis

The slope of the regression line is "b", and the intercept value of regression line is "a"(the value of y when x = 0).

Linear Regression Formula

Formula used for linear regressions is, **y = a + bx**

Intercept value, a, and slope of the line, b, are evaluated using the formulas given below:

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

where,

- y is Dependent Variable that Lies along Y-axis
- a is *Y-Intercept*
- b is *Slope of regression line*
- x is Independent Variable that Lies along X-axis

Properties of Linear Regression

In the linear regression line if the regression parameters a_0 and a_1 are defined, the properties are given as below:

→ Linear regression line reduces the sum of squared differences between observed values and predicted values.

→ Linear regression line always passes through the mean of X and Y variable values.

→ Linear regression constant (b_0) is equal to the y-intercept of the linear regression.

→ Linear regression coefficient (b_0) is the slope of the regression line.

What is Linear Regression Used for?

Various uses of Linear Regression are:

→ It is used in market research and study of customer survey results.

→ It is used for studying performance of engine of automobiles.

→ It is used in deciding the effective price of any goods.

→ It is used in astronomy.

Regression Coefficient

Linear regression line, equation:

$Y = B_0 + B_1X$

where,

B_0 is a Constant

B_1 is Regression Coefficient

Here, B_1 is the regression coefficient and its formula is,

$$\boxed{B_1 = b_1 = \Sigma\,[\,(x_i - x)(y_i - y)\,]\,/\,\Sigma\,[(x_i - x)^2]}$$

where,

→ x_i and y_i are Observed Data Sets

→ x and y are Mean Value

Question 1: Find the linear regression equation for the given data:

x	y
3	8
9	6
5	4
3	2

Regression Coefficient

Linear regression line, equation:

$Y = B_0 + B_1X$

where,

B_0 is a Constant

B_1 is Regression Coefficient

Here, B_1 is the regression coefficient and its formula is,

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where,

→ x_i and y_i are Observed Data Sets

→ x and y are Mean Value

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Calculating intercept and slope value.

x	y	x^2	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6
$\sum x = 20$	$\sum y = 20$	$\sum x^2 = 124$	$\sum xy = 104$

x	y	x^2	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6
$\sum x = 20$	$\sum y = 20$	$\sum x^2 = 124$	$\sum xy = 104$

Using formula for *a*,

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

a = {20 (124) – 20 (104)} / {4 (124) – 400}

a = 400/96 = 4.17

Using formula for *b*,

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

b = {4 (104) – 20 (20)} / {4 (124) – 400}

b = 16/96 = 0.166

So, linear regression equation is:

y=a+bx => y = 4.17 + 0.166x

Find the intercept of linear regression line if $\sum x = 25$, $\sum y = 20$, $\sum x^2 = 90$, $\sum xy = 150$ and n = 5.

Using formula,

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

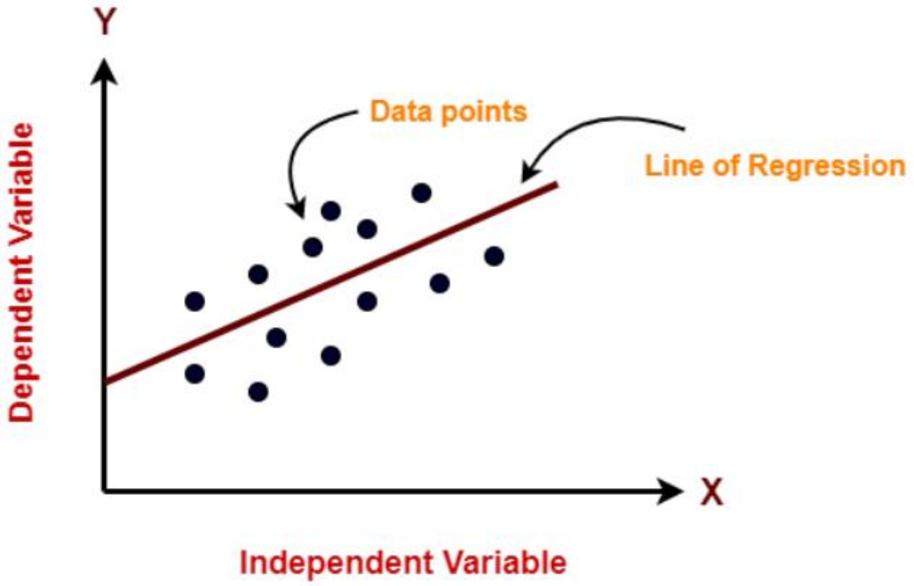
= (20 (90) – 25 (150)) / (5 (90) – 625)

= -1950/-175

= 11.14

SLOPE??

11/11/2024



The equation for a simple linear regression model can be written as follows:

$$y = b_0 + b_1 * x$$

Here, **y is the dependent variable (the variable we are trying to predict)**, **x is the independent variable (the predictor or feature)**.

b0 is the intercept term (the value of y when x is zero), and

b1 is the slope coefficient (the change in y for a unit change in x).

The goal of Linear Regression is to find the best values for b0 and b1 such that the line best fits the data points, minimizing the errors or the difference between the predicted values and the actual values.

Types of Linear Regression?

There are two main types of Linear Regression models:

-- **Simple Linear Regression** and

-- Multiple Linear Regression.

Simple Linear Regression: In simple linear regression, there is only one independent variable (also known as the predictor or feature) and one dependent variable (also known as the response variable). The goal of simple linear regression is to find the best-fitting line to describe the relationship between the independent and dependent variable. The equation for a simple linear regression model can be written as:

$$Y = b_0 + b_1 * X$$

Here, Y is the dependent variable, X is the independent variable, b0 is the intercept term, and b1 is the slope coefficient.

Multiple Linear Regression: In multiple linear regression, there are multiple independent variables and one dependent variable. The goal of multiple linear regression is to find the best-fitting line to describe the relationship between the independent variables and the dependent variable. The equation for a multiple linear regression model can be written as:

$$Y = b_0 + b_1 * X_1 + b_2 * X_2 + \ldots + b_n * X_n$$

Here, Y is the dependent variable, X1, X2, …,

Xn are the independent variables,

b0 is the intercept term, and

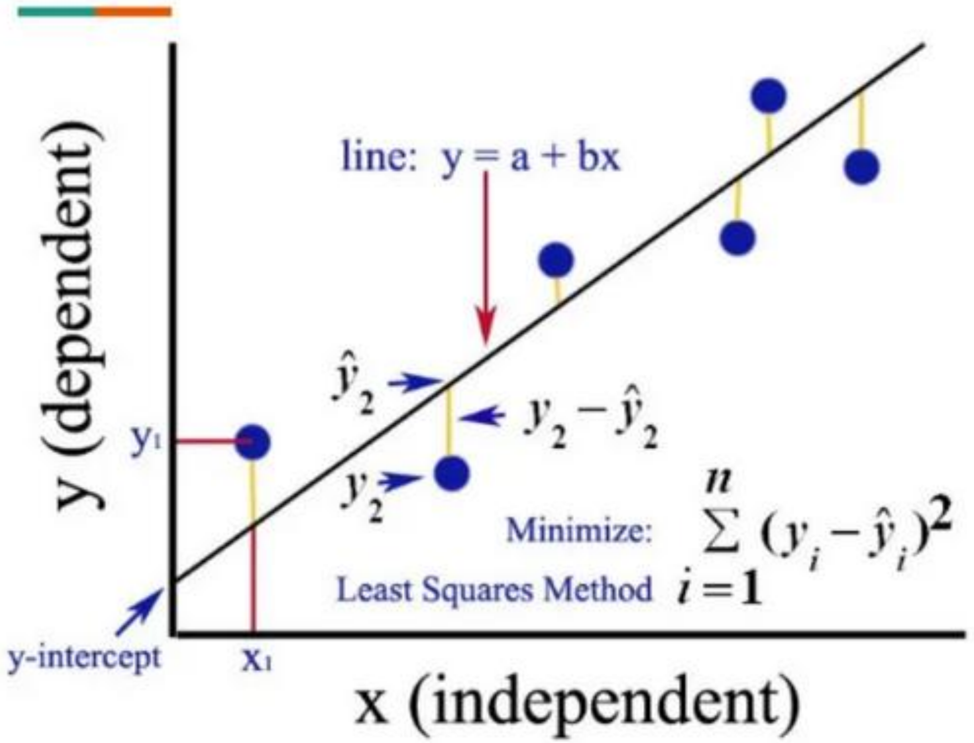
b1, b2, …, bn are the slope coefficients.

In both types of linear regression, the goal is to find the best values for the intercept and slope coefficients that minimize the difference between the predicted values and the actual values.

Finding the best fit line

In machine learning, finding the best-fitting line is crucial in linear regression, as it determines the accuracy of the predictions made by the model. *The best-fitting line is the line that has the smallest difference between the predicted values and the actual values*.

To find the best-fitting line in a linear regression model, one uses a process called "**ordinary least squares (OLS) regression**".
This process involves calculating the sum of the squared differences between the predicted values and the actual values for each data point, and then finding the line that minimizes this sum of squared errors.



The best-fitting line is found by minimizing the residual sum of squares (RSS), *which is the sum of the squared differences between the predicted values and the actual values*.

This is achieved by adjusting the values of the intercept and slope coefficients, also known as c and m, respectively.

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

Once the values of c and m are determined, we can use the linear regression equation to make predictions for new data points. The equation for a simple linear regression model can be written as:

y = c + m * x

Here, y is the dependent variable (the variable we are trying to predict), x is the independent variable (the predictor or feature), c is the intercept term (the value of y when x is zero), and m is the slope coefficient (the change in y for a unit change in x).

Cost function

The cost function, also known as the loss function or objective function, is a measure of how well the model is performing. In linear regression, the cost function is used to calculate the difference between the predicted values and the actual values, also known as the residuals or errors.

The goal of linear regression is to minimize the cost function, which is achieved by finding the best-fitting line that minimizes the sum of the squared differences between the predicted values and the actual values. The most commonly used cost function in linear regression is the **Mean Squared Error (MSE)**, which is calculated as the average of the squared differences between the predicted and actual values:

$$MSE = (1/n) * \Sigma(yi - \hat{y}i)^2$$

* **Here, n is the number of data points, y_i is the actual value, and $\hat{y}_i$ is the predicted value.**

Model Performance

Model performance in linear regression can be evaluated using various metrics that measure how well the model fits the data and how well it generalizes to new data. Go to next slides for common metrics used in linear regression: ………………………………

Example:

Let, there is a linear regression model that predicts housing prices based on the size of the house. Here, 10 data points with the following **actual prices** (y) and **predicted prices** (y_hat):

y = [100, 150, 200, 250, 300, 350, 400, 450, 500, 550]

y_hat = [110, 140, 180, 240, 290, 320, 380, 420, 480, 520]

*** y_mean = sum(y) / len(y) = 325**

1. Sum of Squares Regression (SSR):

SSR is calculated by taking the sum of the squared differences between the predicted values and the mean of the dependent variable. It represents the variability in the dependent variable that is explained by the independent variables. Mathematically, it can be expressed as:

SSR = sum((y_hat — y_mean)^2) = 28300

2. Sum of Squares Error (SSE):

SSE is calculated by taking the sum of the squared differences between the predicted values and the actual values of the dependent variable. Mathematically, it can be expressed as:

$$SSE = sum((y — y\_hat)^2) = 26700$$

3. Mean Squared Error (MSE):

This metric measures the average squared difference between the predicted and actual values. It is calculated as the sum of squared residuals divided by the number of data points. A lower MSE indicates better performance. Mathematically, it can be expressed as:

$$MSE = (1/n) * sum((y — y\_hat)^2) = 2670$$

4. Root Mean Squared Error (RMSE):

This metric measures the square root of the MSE and is useful for interpreting the errors in the same units as the dependent variable. Mathematically, it can be expressed as:

RMSE = sqrt(MSE) = sqrt(2670) = 51.68

5. Mean Absolute Error (MAE):

This metric measures the average absolute difference between the predicted and actual values. It is less sensitive to outliers than MSE, but still provides a measure of the model's accuracy. Mathematically, it can be expressed as:

MAE = (1/n) * sum(abs(y — y_hat)) = 36

6. Sum Of Squares Total (SST):

SST is calculated by taking the sum of the squared differences between the actual values of the dependent variable and its mean value. Mathematically, it can be expressed as:

SST = sum((y — y_mean)^2) = 55000

R-Squared (R^2):

This metric measures the proportion of variance in the dependent variable that is explained by the independent variable(s).

It ranges from 0 to 1, with a higher value indicating better performance.

$$R^2 = SSR / SST = 28300 / 55000 = 0.5164$$

$$R^2 = 1 — SSE / SST = 1 —26700 / 55000 = 0.5164$$

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As one did when studying regression:
let's begin by thinking about the type of problems are trying to solve.
Here are a few examples of classification tasks:

1. Predicting whether a given email is spam.
2. Predicting the type of object in an image.
3. Predicting whether a manufactured good is defective. The point of classification is hopefully clear: we're trying to identify the most appropriate class for an input data point.

Definition 3.1.1 (Classification): A set of problems that seeks to make predictions about unobserved target classes given observed input variables.

Contents

What is Classification?

- Classification is the task of “classifying things” into sub-categories. But, by a machine! If that doesn’t sound like much, imagine your computer being able to differentiate between you and a stranger. Between a potato and a tomato. Between an A grade and an F. Now, it sounds interesting now. In Machine Learning and Statistics, Classification is the problem of identifying to which of a set of categories (subpopulations), a new observation belongs, on the basis of a training set of data containing observations and whose categories membership is known.
- IT is a supervised learning approach.

label

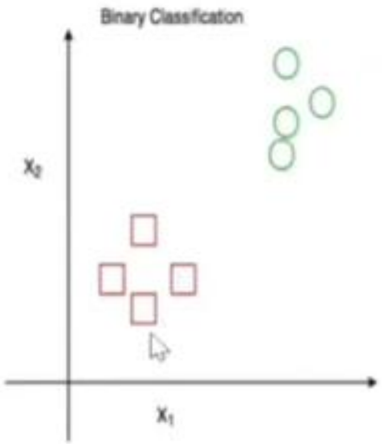
NAME	AGE	GENDER	CLASS OF SPORTS
Ajay	32	0	Football
Mark	40	0	Neither
Sara	16	1	Cricket
Zaira	34	1	Cricket
Sachin	55	0	Neither
Rahul	40	0	Cricket
Pooja	20	1	Neither
Smith	15	0	Cricket
Laxmi	55	1	Football
Michael	15	0	Football

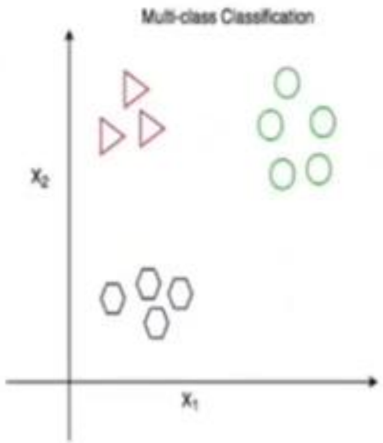
If it exists then one can claim it is supervised

If it doesn't exist then one can claim it is un-supervised

Types of Classification

- Classification is of two types:
- **Binary Classification**: When we have to categorize given data into 2 distinct classes. Example – On the basis of given health conditions of a person, we have to determine whether the person has a certain disease or not.
- **Multiclass Classification**: The number of classes is more than 2. For Example – On the basis of data about different species of flowers, we have to determine which specie our observation belongs.





Classification (ML)

For examplee, there is a review system.

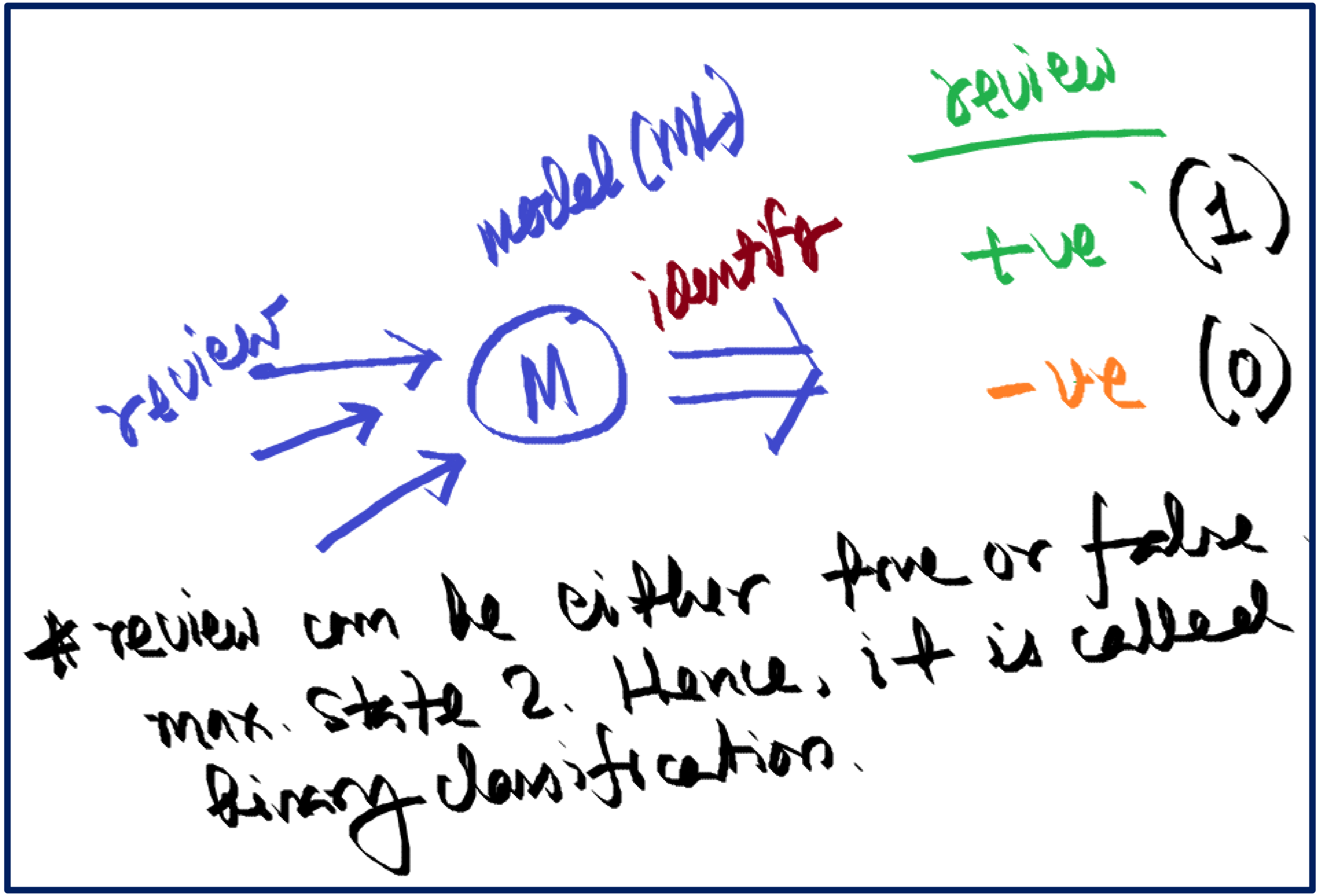
DA course is helpfull.

(T) — yes, it is hellpfull (*5) → (+ve)

(F) — no, it is not supporting (*2) → (-ve)

How will one automates the last slide problem?

It needs a MODEL



* how can M(model) will say the review is true/false.

→ model training data, test data

millions of review → M

class label

For example →

R1 → +ve

R2 → +ve

R3 → -ve

R4 → +ve

...

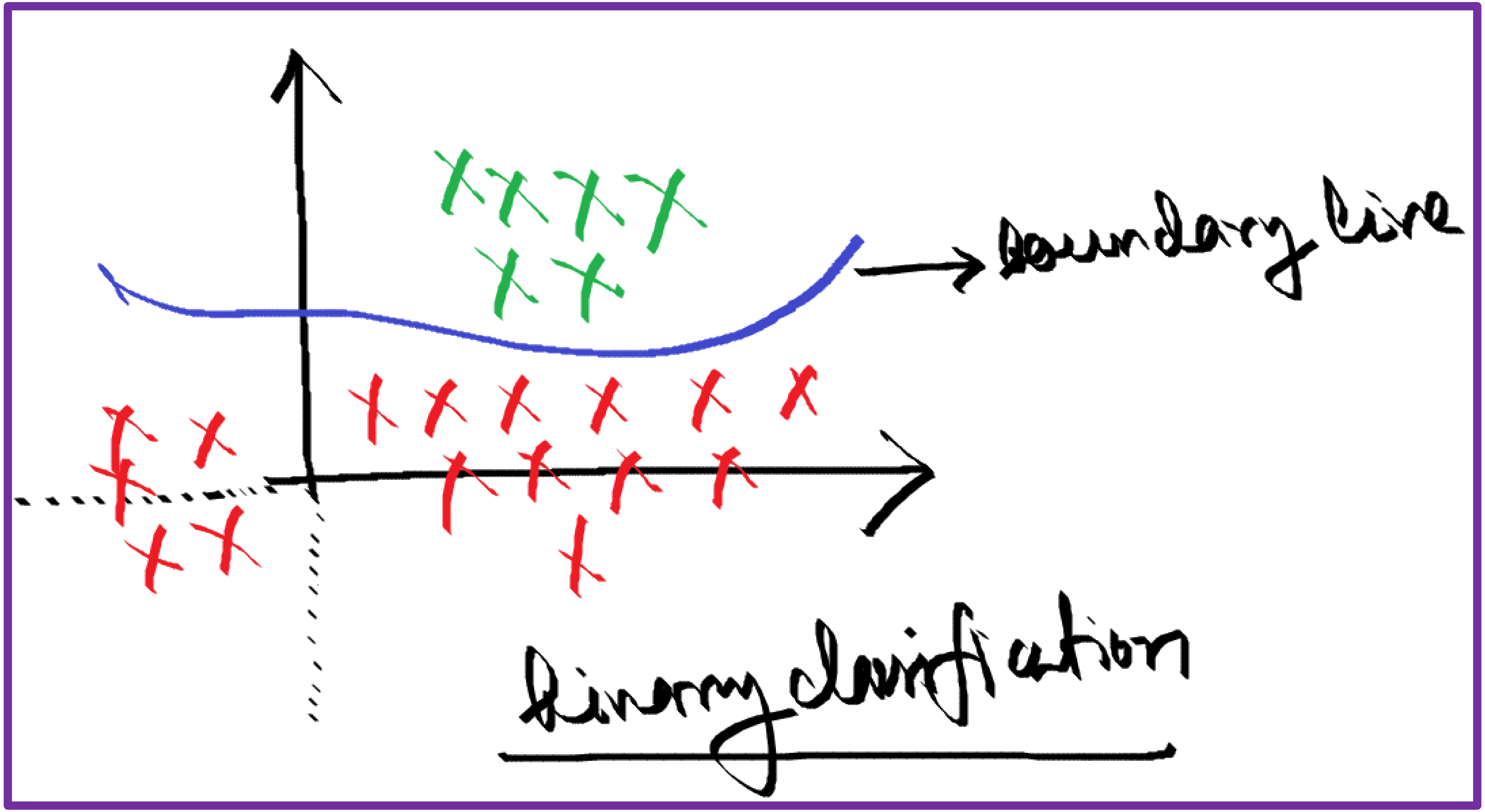
Rn → -ve

add numerical value ...

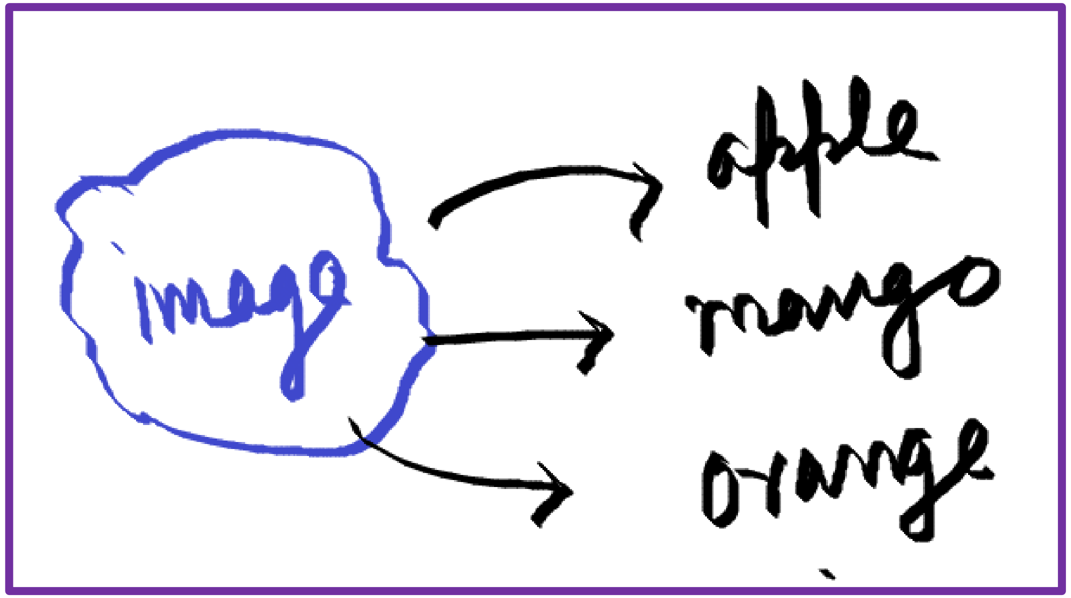
say

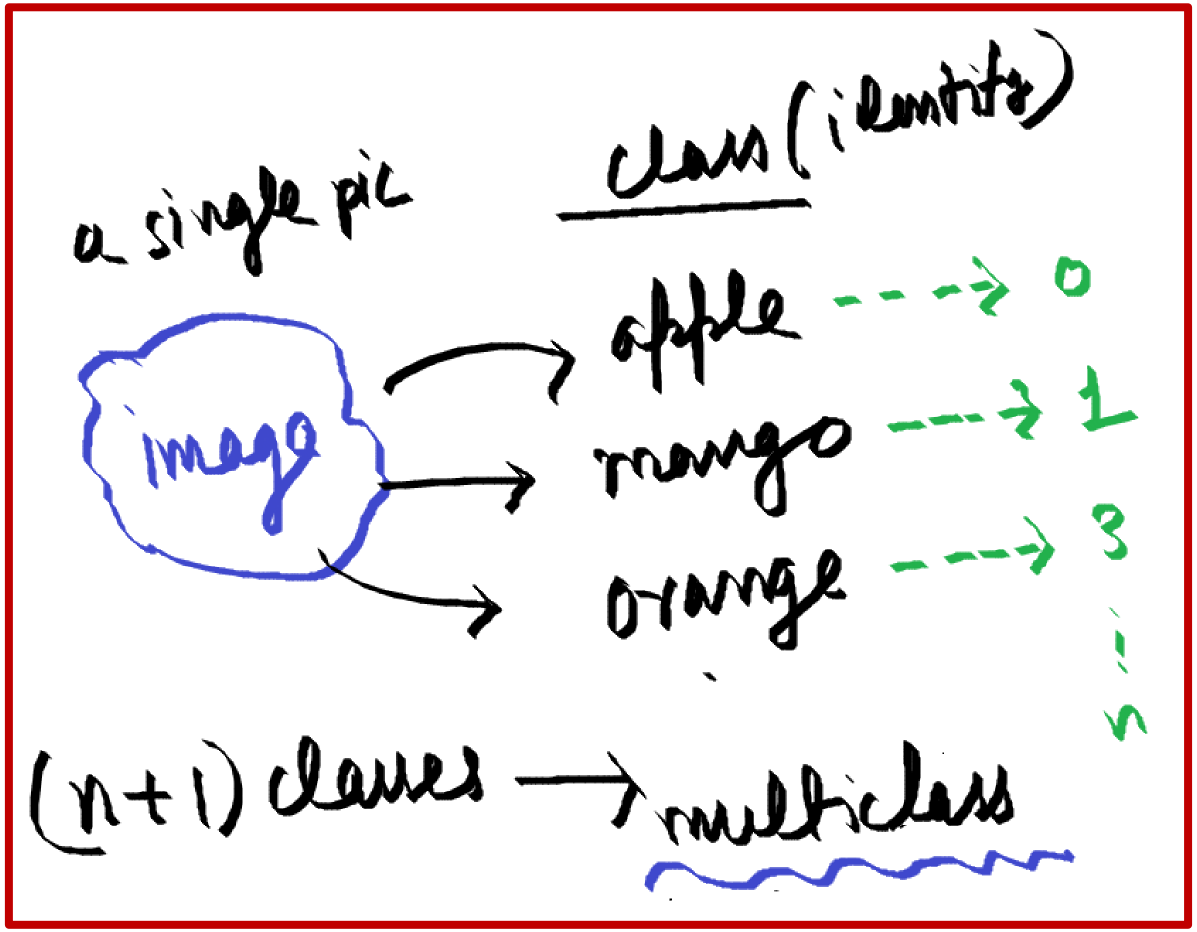
$\geq 3 \rightarrow +ve$

$\leq 2 \rightarrow -ve$


boundary line
binary classification

Multiclass Classification





1.What is Learner?

2.Lazy Learner with example diagram.

3.Eager Learner with example diagram.

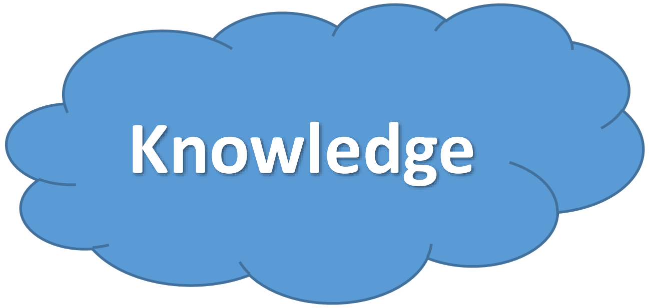
4.Real life example.

5. Lazy Learner vs Eager Learner.

Learner

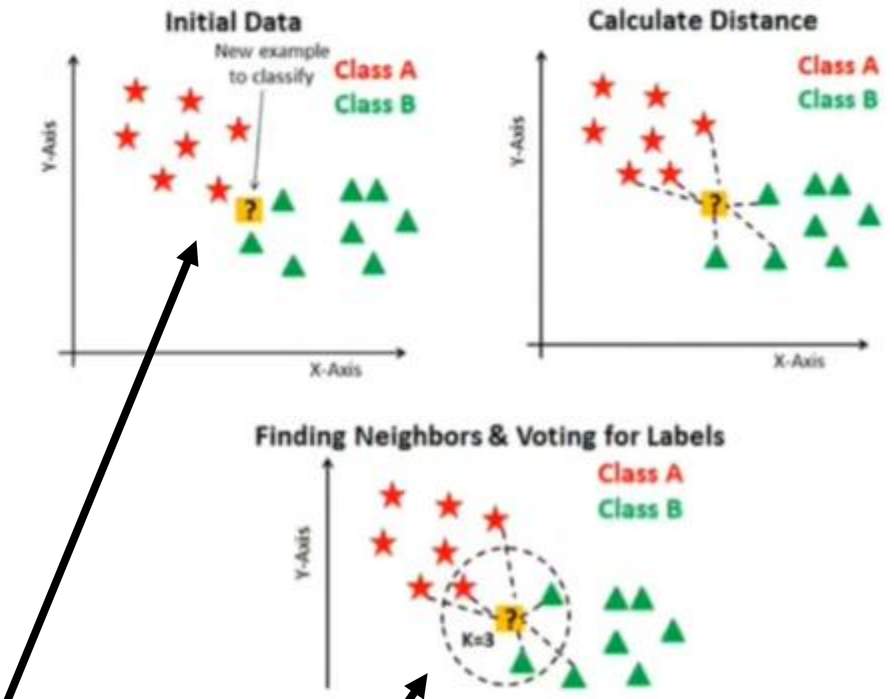
- A Learner or Machine Learning Algorithm is **the program used to learn a machine learning model from data**. Another name is "inducer" (e.g. "tree inducer"). A Machine Learning Model is the learned program that maps inputs to predictions.

- There are two types of learners in classification as lazy learners and eager learners.
- 1.Lazy Leaner
- 2.Eager Learner



Lazy learners simply store the training data and wait until a testing data appear. When it does, classification is conducted based on the most related data in the stored training data. Compared to eager learners, lazy learners have less training time but more time in predicting.

- *Ex. k-nearest neighbor, Case-based reasoning*



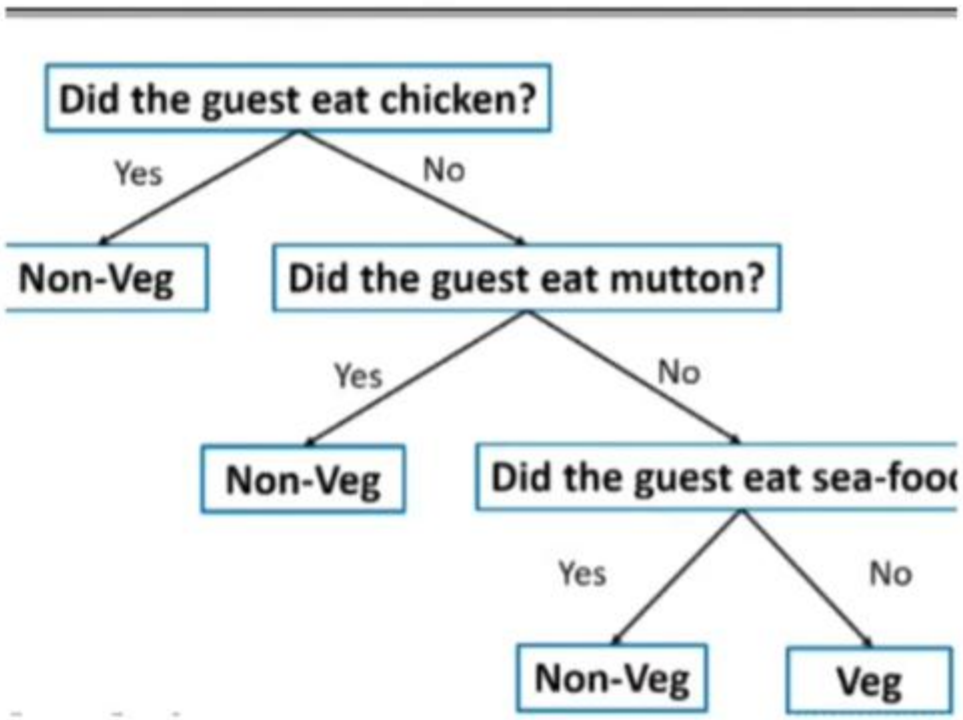
Initially, store data (no activities)

Less training time: take few samples to calculate

More prediction time. Due to distance calculation

Eager Learners develop a classification model based on a training dataset before receiving a test dataset. Opposite to Lazy learners, Eager Learner takes more time in learning, and less time in prediction.

Example: Decision Trees, Naïve Bayes, ANN.



More training time: full data set

less prediction time. Due to created table or tree

Lazy Learner vs Eager Learner

Lazy learner:

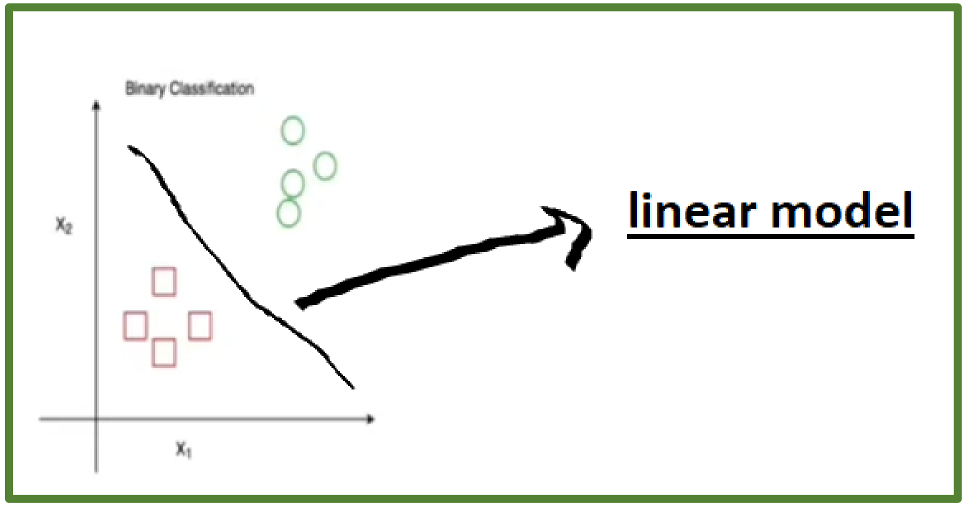
- Just store Data set without learning from it
- Start classifying data when it receive Test data
- So it takes less time learning and more time classifying data
-

Eager learner:

- When it receive data set it starts classifying (learning)
- Then it does not wait for test data to learn
- So it takes long time learning and less time classifying data

Types of ML Classification Algorithms:

- Classification Algorithms can be further divided into the Mainly two category:
- **Linear Models**
  - Logistic Regression
  - Support Vector Machines
- **Non-linear Models**
  - K-Nearest Neighbours
  - Kernel SVM
  - Naïve Bayes
  - Decision Tree Classification
  - Random Forest Classification



Practical Applications of Classification

- 1.Google's self-driving car uses deep learning-enabled classification techniques which enables it to detect and classify obstacles.
- 2.Spam E-mail filtering is one of the most widespread and well-recognized uses of Classification techniques.
- 3.Detecting Health Problems, Facial Recognition, Speech Recognition, Object Detection, and Sentiment Analysis all use Classification at their core.
- 4.Speech Recognition
- 5.Identifications of Cancer tumor cells.
- 6.Drugs Classification
- 7.Biometric Identification, etc.

Clustering in Machine Learning

- Clustering or cluster analysis is a machine learning technique, which groups the unlabelled dataset. It can be defined as ***"A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group."***

Example

Clustering in Machine Learning

- It does it by finding some similar patterns in the unlabeled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns.

- It is an unsupervised learning
- method, hence no supervision is provided to the algorithm, and it deals with the unlabeled dataset.
- Note: Clustering is somewhere similar to the classification algorithm
- , but the difference is the type of dataset that we are using. In classification, we work with the labeled data set, whereas in clustering, we work with the unlabelled dataset.

Unlabeled Data vs Labeled Data

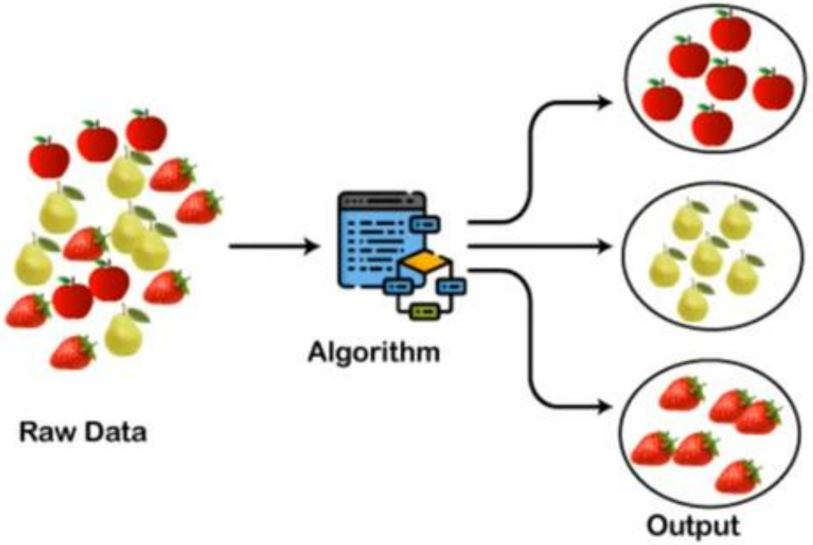
	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
6	4.6	3.4	1.4	0.3	setosa
7	5.0	3.4	1.5	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
9	4.9	3.1	1.5	0.1	setosa

The use of Clustering

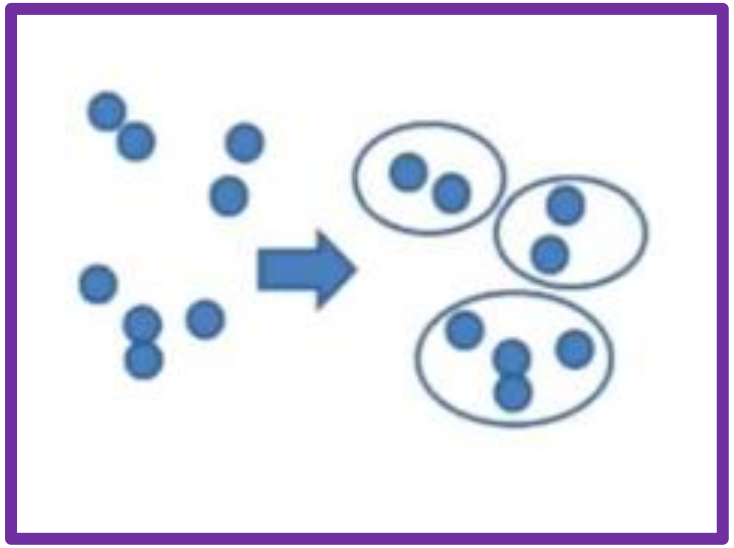
- The clustering technique can be widely used in various tasks. Some most common uses of this technique are:
- Market Segmentation
- Statistical data analysis
- Social network analysis
- Image segmentation
- Anomaly detection, etc.
- Apart from these general usages, it is used by the **Amazon** in its recommendation system to provide the recommendations as per the past search of products. **Netflix** also uses this technique to recommend the movies and web-series to its users as per the watch history.

Clustering Algorithm

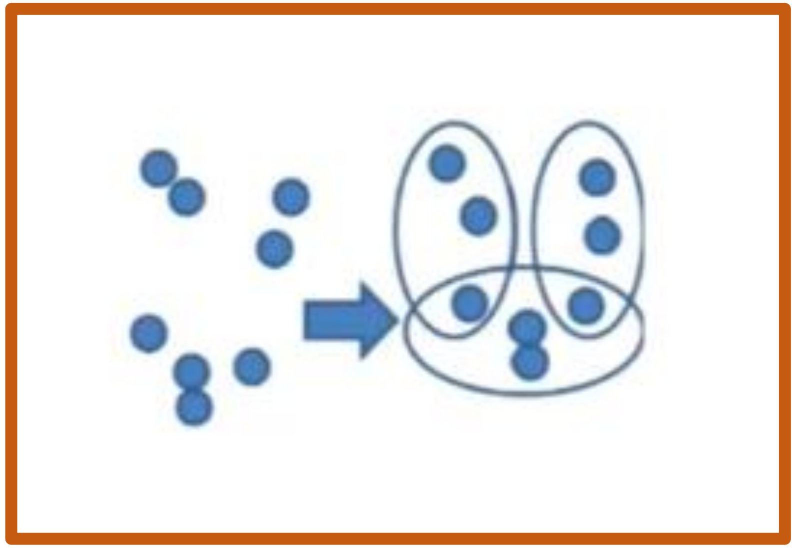


Types of Clustering Methods

- The clustering methods are broadly divided into **Hard clustering** (datapoint belongs to only one group) and **Soft Clustering** (data points can belong to another group also). But there are also other various approaches of Clustering exist. Below are the main clustering methods used in Machine learning:
- **Partitioning Clustering**
- **Density-Based Clustering**
- **Distribution Model-Based Clustering**
- **Hierarchical Clustering**
- **Fuzzy Clustering**

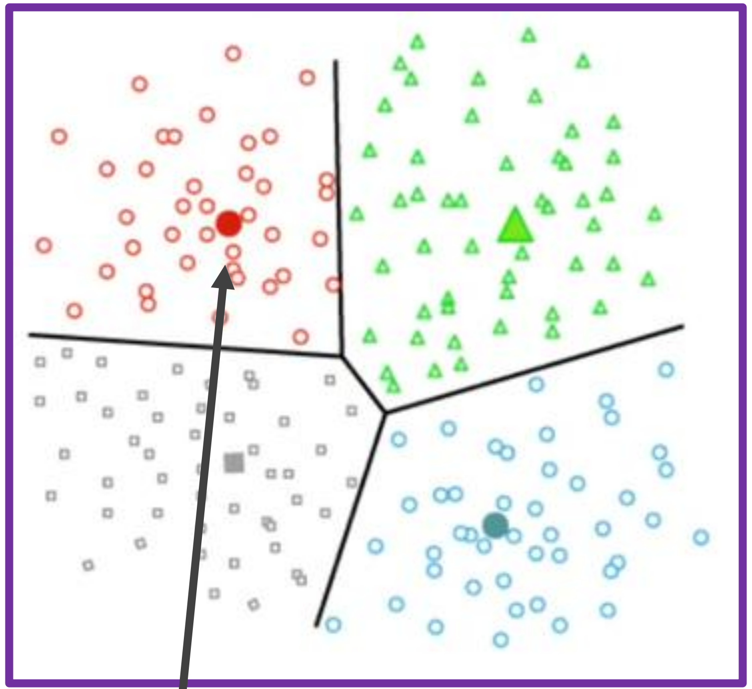


Hard clustering



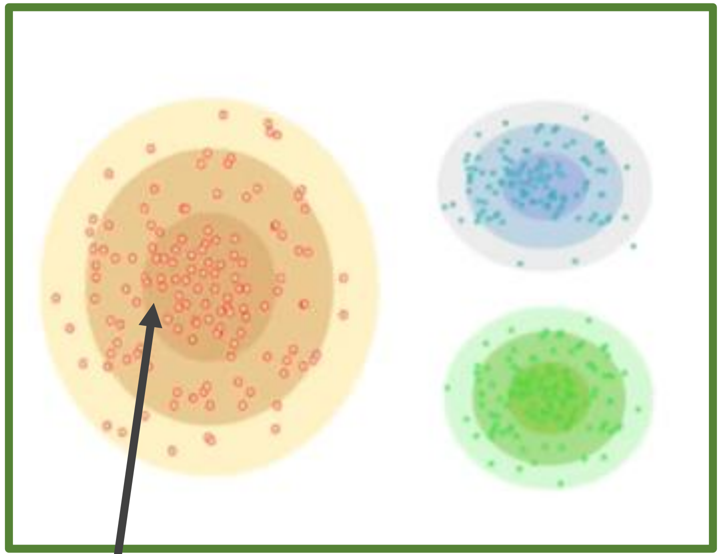
Soft clustering

Partitioning clustering



Centered base: properties or objects are close to center

Dense base clustering



Density based clustering

Applications of Clustering

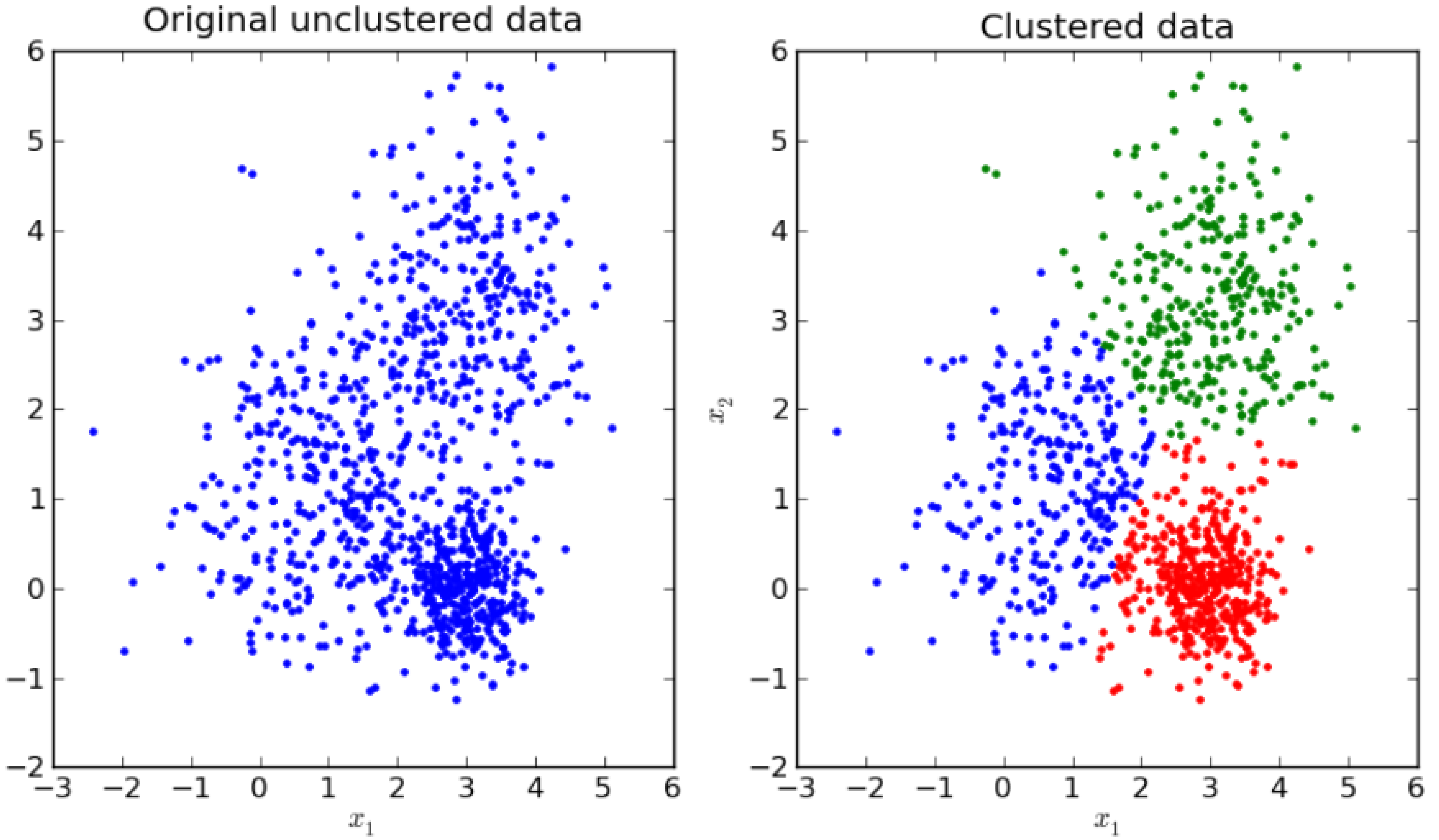
- **In Identification of Cancer Cells:** The clustering algorithms are widely used for the identification of cancerous cells. It divides the cancerous and non-cancerous data sets into different groups.
- **In Search Engines:** Search engines also work on the clustering technique. The search result appears based on the closest object to the search query. It does it by grouping similar data objects in one group that is far from the other dissimilar objects. The accurate result of a query depends on the quality of the clustering algorithm used.
- **Customer Segmentation:** It is used in market research to segment the customers based on their choice and preferences.
- **In Biology:** It is used in the biology stream to classify different species of plants and animals using the image recognition technique.
- **In Land Use:** The clustering technique is used in identifying the area of similar lands use in the GIS database. This can be very useful to find that for what purpose the particular land should be used, that means for which purpose it is more suitable.

18-11-2024

K-means Clustering

- What is clustering? [done]
- Why would we want to cluster?
- How would you determine clusters?
- How can you do this efficiently?

K-means clustering



The key objective of a k-means algorithm is to organize data into clusters such that there is high intra-cluster similarity and low inter-cluster similarity.
An item will only belong to one cluster, not several, that is, it generates a specific number of disjoint, non-hierarchical clusters.

K-means uses the strategy of divide and concur, and it is a classic example for an expectation maximization (EM) algorithm.
EM algorithms are made up of two steps:

– **The first step** is known as expectation(E) and is used to find the expected point associated with a cluster; and

– **The second step** is known as maximization(M) and is used to improve the estimation of the cluster using knowledge from the first step.

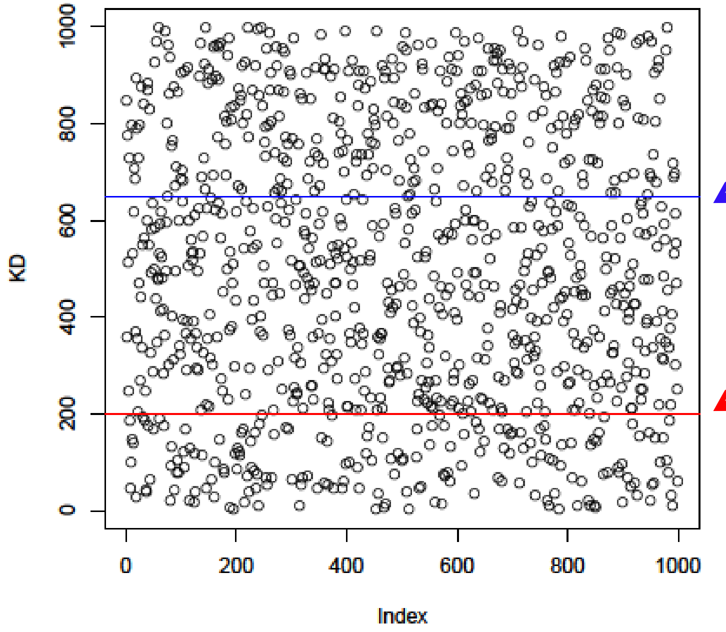
K-means Clustering

- Strengths
  - Simple iterative method
  - User provides “K”
- Weaknesses
  - Often too simple → bad results
  - Difficult to guess the correct “K”

K-means Clustering

Basic Algorithm:

- Step 0: select K
- Step 1: randomly select initial cluster seeds



Seed 1
650

Seed 2
200

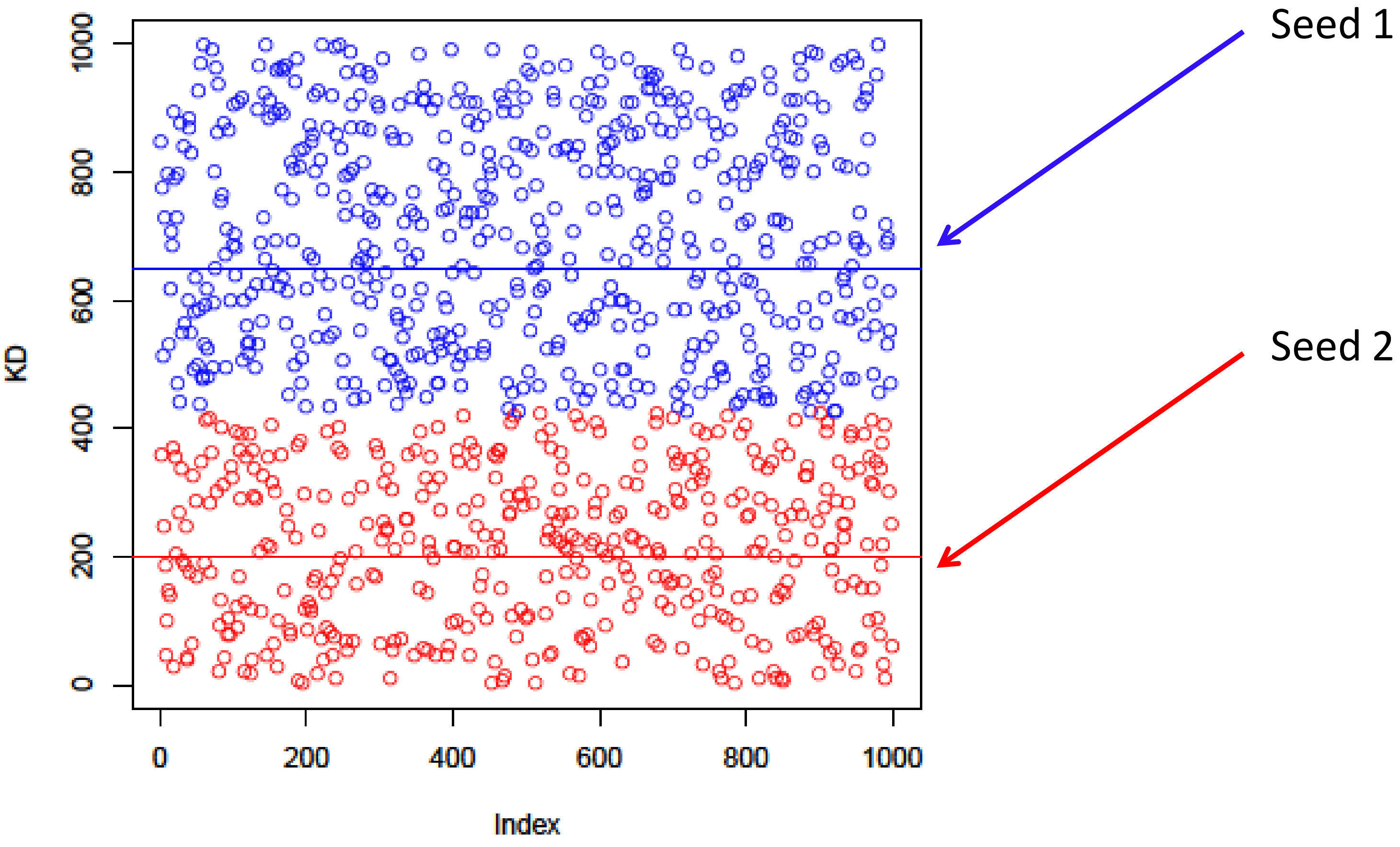
K-means Clustering

- An initial cluster seed represents the “mean value” of its cluster.
- In the preceding figure:
  - Cluster seed 1 = 650
  - Cluster seed 2 = 200

K-means Clustering

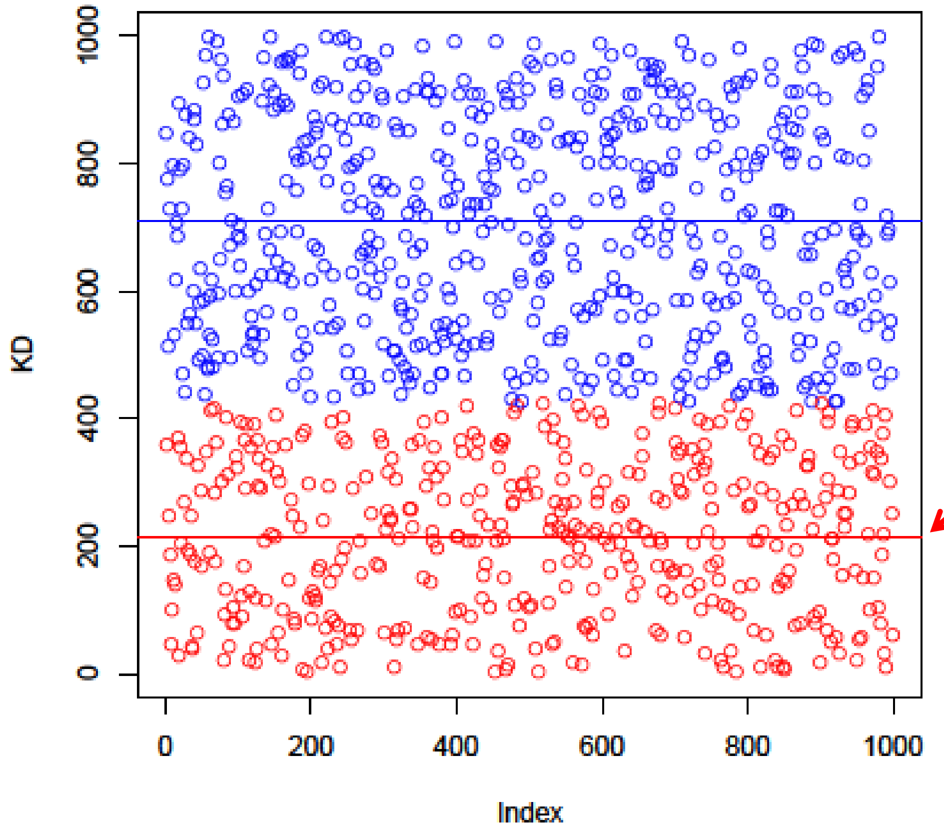
- Step 2: calculate distance from each object to each cluster seed.
- What type of distance should we use?
  - **Squared Euclidean distance**
- Step 3: Assign each object to the closest cluster

K-means Clustering



K-means Clustering

- Step 4: Compute the new centroid for each cluster



Cluster Seed 1
708.9

Cluster Seed 2
214.2

K-means Clustering

- Iterate:
  - Calculate distance from objects to cluster centroids.
  - Assign objects to closest cluster
  - Recalculate new centroids
- Stop based on convergence criteria
  - No change in clusters
  - Max iterations

K-means Issues

- Distance measure is squared Euclidean
  - Scale should be similar in all dimensions
    - Rescale data?
  - Not good for nominal data. Why?
- Approach tries to minimize the within-cluster sum of squares error (WCSS)
  - Implicit assumption that SSE is similar for each group

WCSS

- The over all WCSS is given by: $\sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$
- The goal is to find the smallest WCSS
- Does this depend on the initial seed values?
- Possibly.

Bottom Line

- K-means
  - Easy to use
  - Need to know K
  - May need to scale data
  - Good initial method
- Local optima
  - No guarantee of optimal solution
  - Repeat with different starting values

K-Means Clustering Example

Distance Formula

- **Solve K-Means algorithm using:**
  Euclidean distance formula
- **Solve K-Means algorithm using:**
  Manhattan distance formula

- Let's consider we have cluster points **P1(1,3) , P2(2,2) , P3(5,8) , P4(8,5) , P5(3,9) , P6(10,7) , P7(3,3) , P8(9,4) , P9(3,7).**
- First, we take our **K value as 3** and we assume that our **Initial cluster centers are P7(3,3), P9(3,7), P8(9,4) as C1, C2, C3.**
- That's means:
- **C1=P7(3,3)=(3,3)**
- **C2=P9(3,7)=(3,7)**
- **C3=P8(9,4)=(9,4)**

Randomly chosen

- Find the distance between data points and Centroids. which data points have a minimum distance that points moved to the nearest cluster centroid.

- **Iteration-01:**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **We calculate the distance of each point from each of the center of the three clusters.**
- C1(3,3) to all data points-------> P1(1,3) , P2(2,2) , P3(5,8) , P4(8,5) , P5(3,9) , P6(10,7) , P7(3,3) , P8(9,4) , P9(3,7).
- C2(3,7) to all data points-------> P1(1,3) , P2(2,2) , P3(5,8) , P4(8,5) , P5(3,9) , P6(10,7) , P7(3,3) , P8(9,4) , P9(3,7).
- C3(9,4) to all data points-------> P1(1,3) , P2(2,2) , P3(5,8) , P4(8,5) , P5(3,9) , P6(10,7) , P7(3,3) , P8(9,4) , P9(3,7).

❑ Calculate the distance between data points and K (C1,C2,C3)

❖ For P1

- C1P1 =>(3,3)(1,3) => sqrt[(1-3)²+(3-3)²] => sqrt[4] =>2
- C2P1 =>(3,7)(1,3)=> sqrt[(1-3)²+(3-7)²] => sqrt[20] =>4.5
- C3P1 =>(9,4)(1,3) => sqrt[(1-9)²+(3-4)²]=> sqrt[65] =>8.1

❖ For P2

- C1P2=> (3,3)(2,2) => sqrt[(2-3)²+(2-3)²] => sqrt[2] =>1.4
- C2P2 =>(3,7)(2,2)=> sqrt[(2-3)²+(2-7)²] => sqrt[26] =>5.1
- C3P2 =>(9,4)(2,2) => sqrt[(2-9)²+(2-4)²]=> sqrt[53] =>7.3

❖ For P3,

- C1P2 =>(3,3)(5,8) => sqrt[(5-3)²+(8-3)²] => sqrt[29] =>5.3
- C2P2 =>(3,7)(5,8)=> sqrt[(5-3)²+(8-7)²] => sqrt[5] =>2.2
- C3P2 =>(9,4)(5,8) => sqrt[(5-9)²+(8-4)²]=> sqrt[32] =>5.7

- Similarly for other distances..

Iteration 1

Data Points	Centroid (3,3)	Centroid (3,7)	Centroid (9,4)	Cluster
P1(1,3)	2	4.5	8.1	C1
P2(2,2)	1.4	5.1	7.3	C1
P3(5,8)	5.3	2.2	5.7	C2
P4(8,5)	5.4	5.4	5.1	C3
P5(3,9)	6	2	7.9	C2
P6(10,7)	8.1	7	3.2	C3
P7(3,3)	0	4	6.1	C1
P8(9,4)	6.1	6.7	0	C3
P9(3,7)	4	0	6.7	C2

- C1P1 =>(3,3)(1,3) => sqrt[(1-3)^2+(3-3)^2] => sqrt[4] =>2

C1P2 =>(3,3)(2,2) => sqrt[(2–3)^2+(2–3)^2] => sqrt[2] =>1.4

C1P2 =>(3,3)(5,8) => sqrt[(5–3)^2+(8–3)^2] => sqrt[29] =>5.3

....

C1P9=>(3,3),(3,7) => sqrt[(3–3)^2+(7–3)^2] => sqrt[16] =>4

....

C2P1 =>(3,7)(1,3)=> sqrt[(1–3)^2+(3–7)^2] => sqrt[20] =>4.5

....

C3P1 =>(9,4)(1,3) => sqrt[(1–9)^2+(3–4)^2]=> sqrt[65] =>8.1

Iteration 1, step-1

- ❑Cluster 1 => P1(1,3) , P2(2,2) , P7(3,3)
- ❑Cluster 2 => P3(5,8) , P5(3,9) , P9(3,7)
- ❑Cluster 3 => P4(8,5) , P6(10,7) , P8(9,4)

- **Now, We re-compute the new clusters and the new cluster center is computed by taking the mean of all the points contained in that particular cluster.**

- ❑New center of Cluster 1 => (1+2+3)/3 , (3+2+3)/3 => (2,2.7)
- ❑New center of Cluster 2 => (5+3+3)/3 , (8+9+7)/3 => (3.7,8)
- ❑New center of Cluster 3 => (8+10+9)/3 , (5+7+4)/3 => (9,5.3)

- **Iteration 1 is over.**

- Iteration 1 is over. Now, let us take our new center points and repeat the same steps which are to calculate the distance between data points and new center points with the Euclidean formula and find cluster groups.

  - New C3= (2,2.7)
  - New C2=(3.7,8)
  - New C1= (9,5.3)

Iteration 2

Data Points	Centroid (2,2.7)	Centroid (3.7,8)	Centroid (9,5.3)	Cluster
P1(1,3)	1.0	4.5	8.3	C1
P2(2,2)	0.7	6.2	7.7	C1
P3(5,8)	6.1	1.3	4.8	C2
P4(8,5)	6.4	5.2	1.0	C3
P5(3,9)	6.4	1.2	7.0	C2
P6(10,7)	9.1	6.4	1.9	C3
P7(3,3)	1.0	5.0	6.4	C1
P8(9,4)	7.1	6.6	1.3	C3
P9(3,7)	4.4	1.2	6.2	C2

Iteration 2, step-2

- Cluster 1 => P1(1,3) , P2(2,2) , P7(3,3)
- Cluster 2 => P3(5,8) , P5(3,9) , P9(3,7)
- Cluster 3 => P4(8,5) , P6(10,7) , P8(9,4)

New Center point :

- Center of Cluster 1 => (1+2+3)/3 , (3+2+3)/3 => 2,2.7
- Center of Cluster 2 => (5+3+3)/3 , (8+9+7)/3 => 3.7,8
- Center of Cluster 3 => (8+10+9)/3 , (5+7+4)/3 => 9,5.3

Iteration-2 is over.

Last step

- We got the same centroid and cluster groups which indicates that this dataset has only 2 groups. K-Means clustering stops iteration because of the same cluster repeating so no need to continue iteration and display the last iteration as the best cluster groups for this dataset.

- **After Iteration 1:**
- New center of Cluster 1 => 2,2.7
- New center of Cluster 2 => 3.7,8
- New center of Cluster 3 => 9,5.3

- **After Iteration 2:**
- Center of Cluster 1 => 2,2.7
- Center of Cluster 2 => 3.7,8
- Center of Cluster 3 => 9,5.3

Answer

After apply K-Means algorithm the clusters from given data point are:

- ❑Cluster 1 => P1(1,3) , P2(2,2) , P7(3,3)
- ❑Cluster 2 => P3(5,8) , P5(3,9) , P9(3,7)
- ❑Cluster 3 => P4(8,5) , P6(10,7) , P8(9,4)

Center point for each cluster given below:

- ❑Center of Cluster 1 => (1+2+3)/3 , (3+2+3)/3 => 2,2.7
- ❑Center of Cluster 2 => (5+3+3)/3 , (8+9+7)/3 => 3.7,8
- ❑Center of Cluster 3 => (8+10+9)/3 , (5+7+4)/3 => 9,5.3

- **Step-01:**
- Choose the number of clusters K.
- **Step-02:**
- Randomly select any K data points as cluster centers.
- **Step-03:**
- Calculate the distance between each data point and each cluster center.
- **Step-04:**
- Assign each data point to some cluster.
- **Step-05:**
- Re-compute the center of newly formed clusters.
- **Step-06:**
- Keep repeating the procedure from Step-03 to Step-05 until any of the following stopping criteria is met-
- Center of newly formed clusters do not change
- Data points remain present in the same cluster
- Maximum number of iterations are reached

Hamming Distance

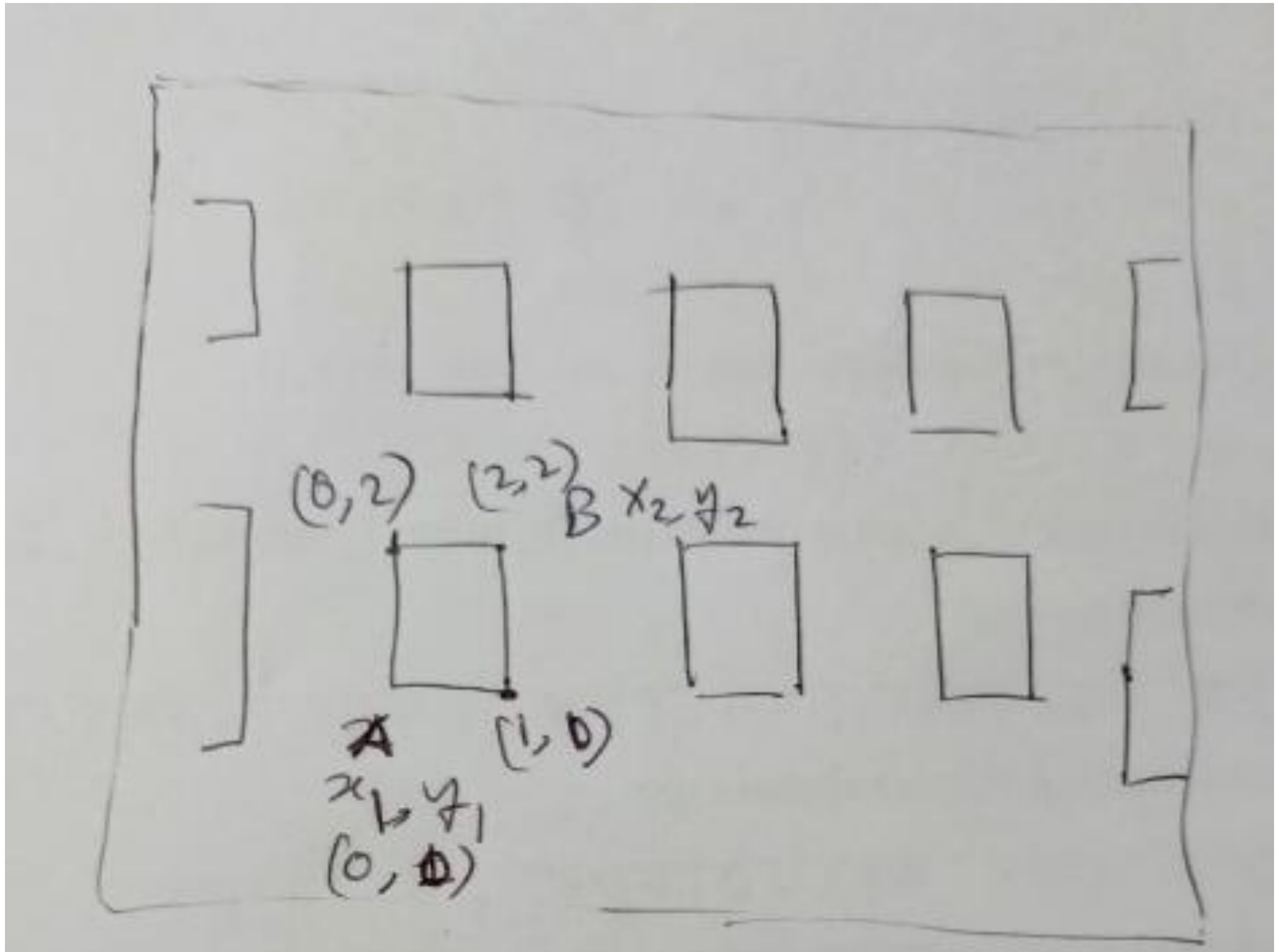
Manhattan distance is also called Hamming distance when all features are binary

– Count the number of difference between two binary vectors

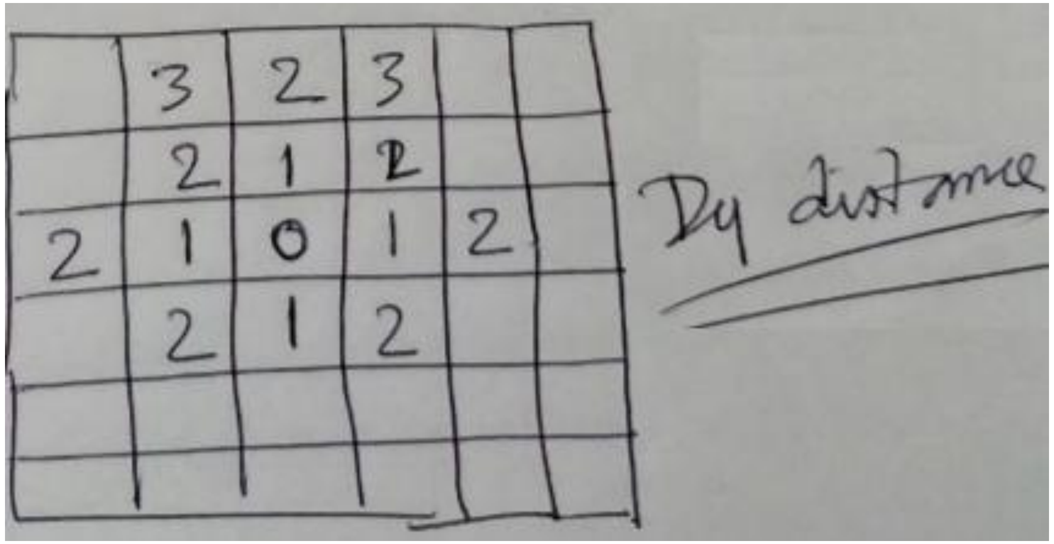
– Example, $x, y \in \{0, 1\}^{17}$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
x	0	1	1	0	0	1	0	0	1	0	0	1	1	1	0	0	1
y	0	1	1	1	0	0	0	0	1	1	1	1	1	1	0	1	1

$$d(x, y) = 5$$



$$d(A,B) = |x_2 - x_1| + |y_2 - y_1|$$
$$= |2-0| + |2-0| = 4$$



Problem

Eight points:

$P_1 \longrightarrow (2,10)$
$P_2 \longrightarrow (2,5)$
$P_3 \longrightarrow (8,4)$
$P_4 \longrightarrow (5,8)$
$P_5 \longrightarrow (7,5)$
$P_6 \longrightarrow (6,4)$
$P_7 \longrightarrow (1,2)$
$P_8 \longrightarrow (4,9)$

Given 3 clusters –

$C_1 \longrightarrow (2,10)$
$C_2 \longrightarrow (5,8)$
$C_3 \longrightarrow (1,2)$

distance formula: $d(a,b) = |x_2 - x_1| + |y_2 - y_1|$

– for example, we will subtract P_1 from C_1, C_2 & C_3 and decide which one is closer.

for example, we will subtract p_i from c_1, c_2 & c_3 and decide which one is closer.

$i = 1$

	$C_1(2,10)$	$C_2(5,8)$	$C_3(1,2)$	selection cluster
$P_1(2,10)$	0	$(5-2)+(8-10)$	$(1-2)+(2-10)$	C_1
$P_2(2,5)$	$(2-2)+(10-5)$	$(5-2)+(8-5)$	$(1-2)+(2-5)$	C_3
$P_3(8,4)$				C_2
$P_4(5,8)$				C_2
$P_5(7,5)$				C_2
$P_6(6,4)$				C_2
$P_7(1,2)$				C_3
$P_8(4,9)$				C_2

* we need recomputation of C_i

$C_1 \rightarrow$ only one point. no change

$C_2 \rightarrow$ ~~P_2~~, P_3, P_4, P_5, P_8 P_6

~~(2,5)~~

(8,4) (5,8) (7,5) (4,9) (6,4)

take all $x \rightarrow$

$$2+8+5+7+4/5 = 30/5 = 6$$

take all $y \rightarrow$

$$4+8+5+9+4/5 = 30/5 = 6$$

$C_2 \rightarrow (6,6)$

same as C_3.

Introduction

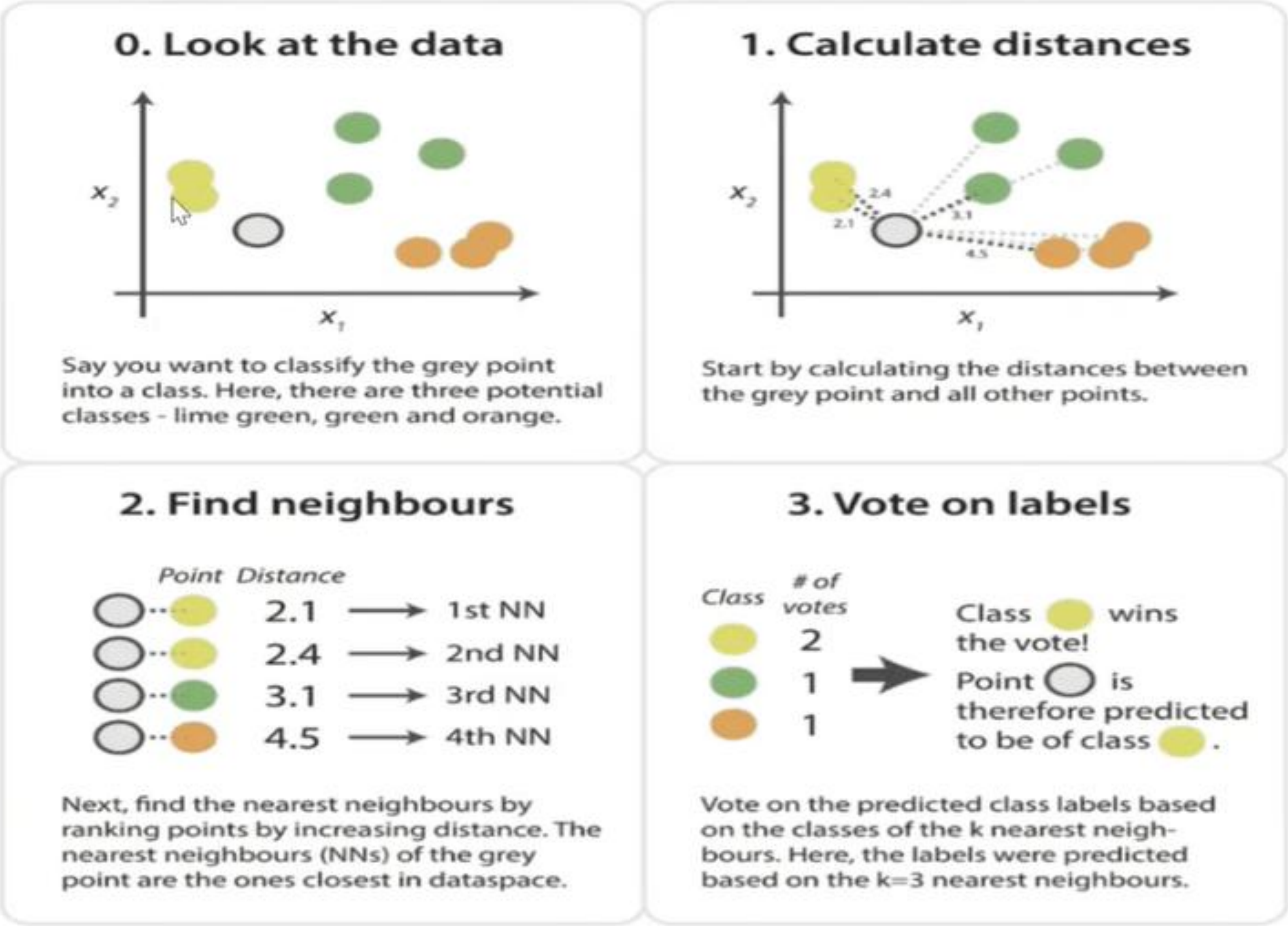
K-nearest neighbors (KNN) is a type of supervised learning algorithm used for both regression and classification. KNN tries to predict the correct class for the test data by calculating the distance between the test data and all the training points. Then select the K number of points which is closest to the test data. The KNN algorithm calculates the probability of the test data belonging to the classes of 'K' training data and class holds the highest probability will be selected. In the case of regression, the value is the mean of the 'K' selected training points.

NAME	AGE	GENDER	CLASS OF SPORTS
Ajay	32	0	Football
Mark	40	0	Neither
Sara	16	1	Cricket
Zaira	34	1	Cricket
Sachin	55	0	Neither
Rahul	40	0	Cricket
Pooja	20	1	Neither
Smith	15	0	Cricket
Laxmi	55	1	Football
Michael	15	0	Football

Here male is denoted with numeric value 0 and female with 1.

Let's find in which class of people Angelina will lie whose k factor is 3 and age is 5

An example to understand



Important Characteristics of KNN-Algorithm

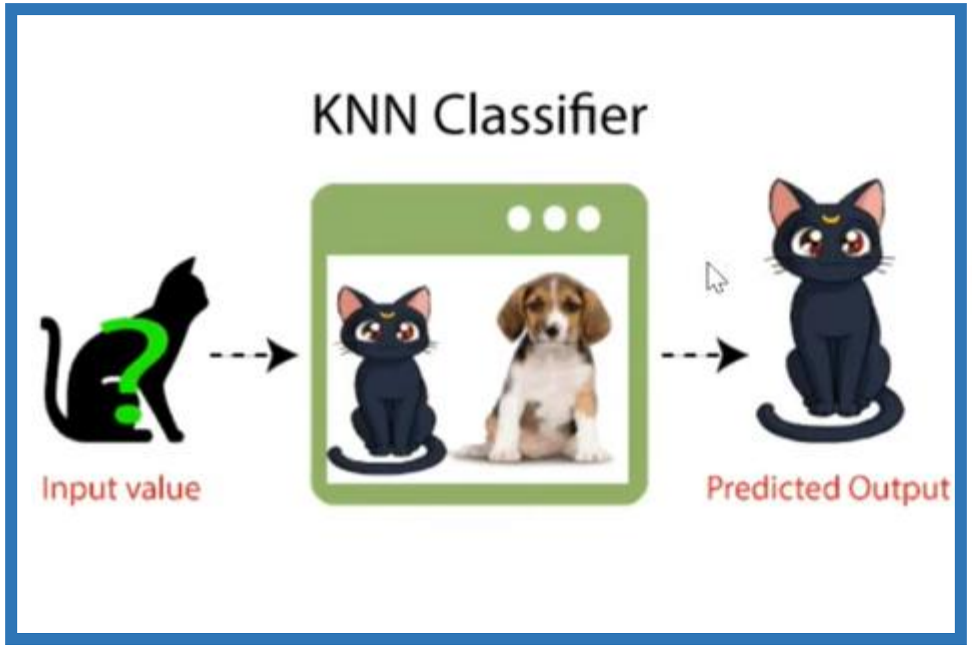
1. K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on Supervised Learning technique.

2. K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.

3. K-NN is a ***non-parametric algorithm***, which means it does not make any assumption on underlying data.

4. It is also called a ***lazy learner algorithm*** because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.

5. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data



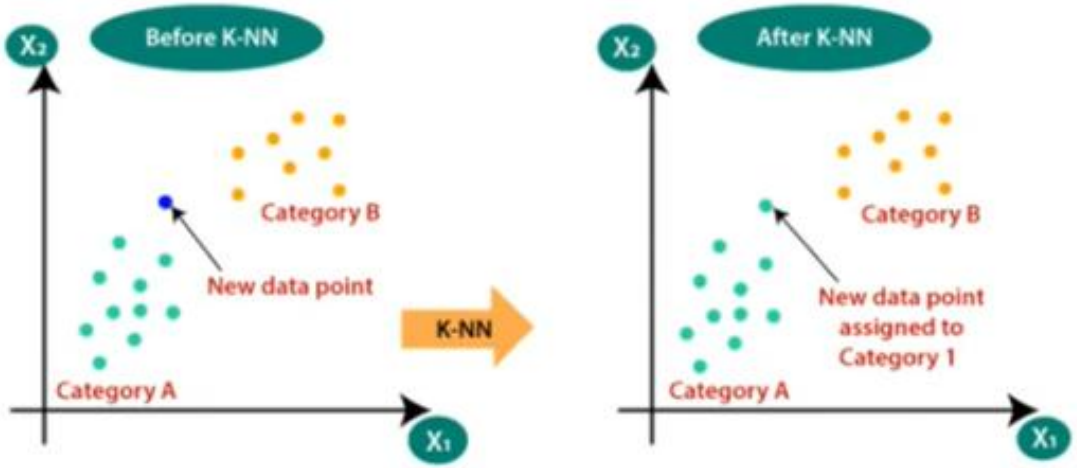
Example

Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

For exmaple, TESLA car: Identify an object?

Why do we need a K-NN Algorithm

Suppose there are two categories, i.e., Category A and Category B, and we have a new data point x1, so this data point will lie in which of these categories. To solve this type of problem, we need a K-NN algorithm. With the help of K-NN, we can easily identify the category or class of a particular dataset. Consider the below diagram:



How does K-NN work?

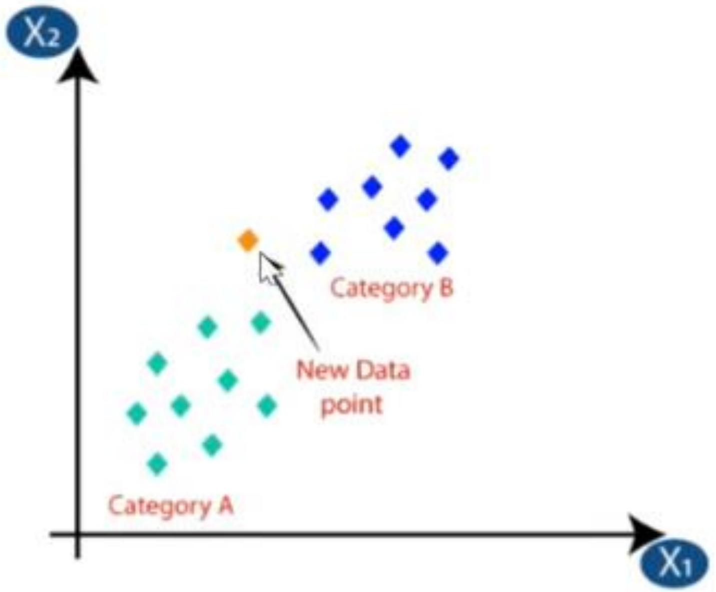
The K-NN working can be explained on the basis of the below algorithm:

- Step-1: Select the number K of the neighbors
- Step-2: Calculate the Euclidean distance of K number of neighbors
- Step-3: Take the K nearest neighbors as per the calculated Euclidean distance.
- Step-4: Among these k neighbors, count the number of the data points in each category.
- Step-5: Assign the new data points to that category for which the number of the neighbor is maximum.
- Step-6: Our model is ready.

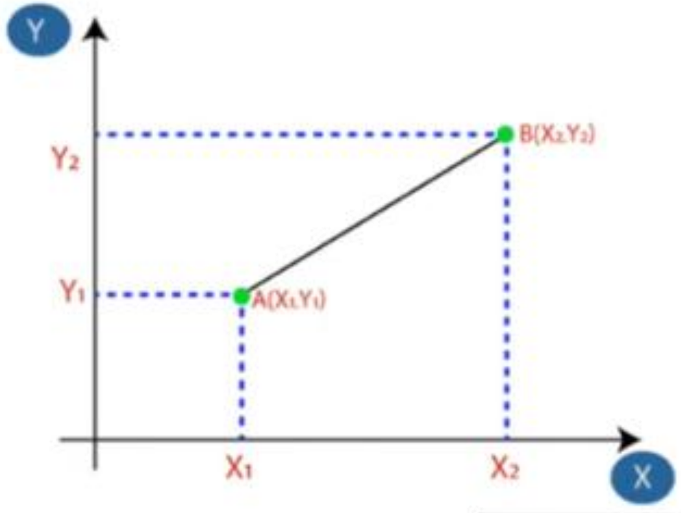
Example

Suppose we have a new data point and we need to put it
in the

required category. Consider the below image:



- Firstly, we will choose the number of neighbors, so we will choos the $k=5$.
- Next, we will calculate the Euclidean distance between the data points. The Euclidean distance is the distance between two points, which we have already studied in geometry. It can be calculated as:



Euclidean Distance between A_1 and $B_2 = \sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$

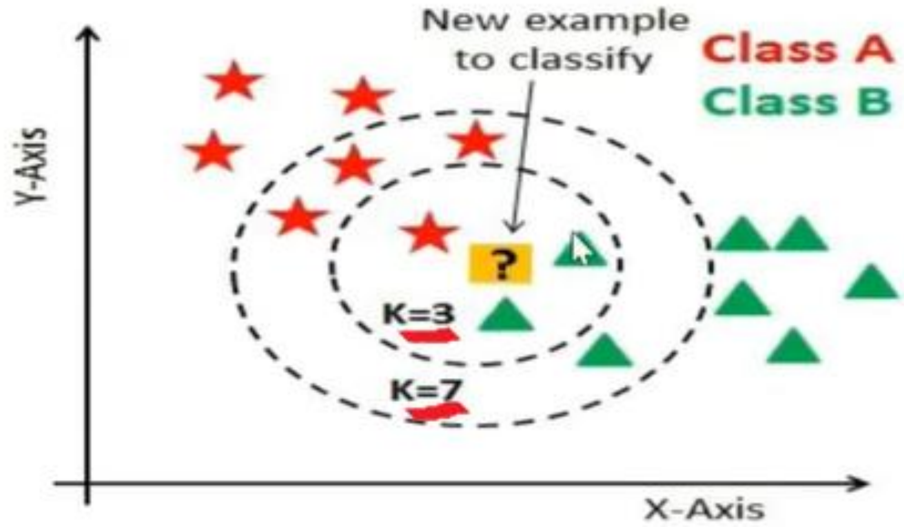
By calculating the Euclidean distance we got the nearest neighbors, as three nearest neighbors in category A and two nearest neighbors in category B. Consider the below image:

As we can see the 3 nearest neighbors are from category A, hence this new data point must belong to category A.

How to select the value of K in the K-NN Algorithm?

Advantages & Disadvantages of KNN Algorithm

Kvalue indicates the count of the nearest neighbors. We have to compute distances between test points and trained labels points. Updating distance metrics with every iteration is computationally expensive, and that's why KNN is a lazy learning algorithm.



- As you can verify from the above image, if we proceed with K=3, then we predict that test input belongs to class B, and if we continue with K=7 then we predict that test input belongs to class A.
- That's how you can imagine that the K value has a powerful effect on KNN performance.

How do we choose K?

Sqrt(n), where n is a total number of data points(if in case n is even we have to make the value odd by adding 1 or subtracting 1 that helps in select better)

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2

57 kg	170 cm	?

$n = 9$

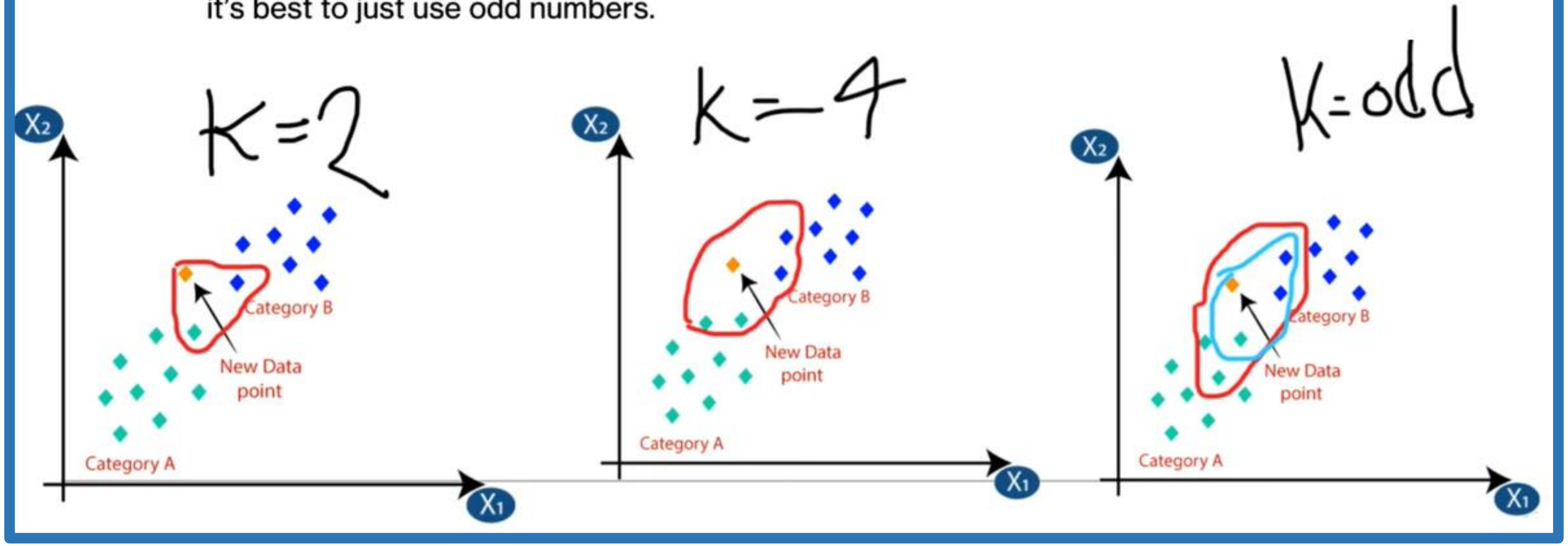
so, $K = \sqrt{9} = 3$

If n=10 then,
make n=10-1=9 or n=10+1=11
so,

when n=9 then k=3
when n=11 then k=sqrt(11)=3.32 =3

K's value should be odd

Just to confirm what others said, *k* is best as an odd value because KNN is a majority voting algorithm. An even number may lead to a tie in samples at the decision boudaries, it's best to just use odd numbers.



Weight(x2)	Height(y2)	Class
51	167	Underweight
62	182	Normal
69	176	Normal
64	173	Normal
65	172	Normal
56	174	Underweight
58	169	Normal
57	173	Normal
55	170	Normal

Example-1

57 kg	170 cm	?

O(x1,y1)
A(51,167)
B(62,182)
C(69,176)
D(64,173)
E(65,172)
F(56,174)
G(58,169)
H(57,173)
I(55,170)

Weight(x2)	Height(y2)	Class
51	167	Underweight
62	182	Normal
69	176	Normal
64	173	Normal
65	172	Normal
56	174	Underweight
58	169	Normal
57	173	Normal
55	170	Normal

57 kg	170 cm	?

Z(x2,y2) or Z(57,170)

Calculate the distance

O(x1,y1)
A(51,167)
B(62,182)
C(69,176)
D(64,173)
E(65,172)
F(56,174)
G(58,169)
H(57,173)
I(55,170)

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2

57 kg	170 cm	?

Z(x2,y2) or Z(57,170)

Looking at the new data, we can consider the last three rows from the table—K=3.

Since the majority of neighbors are classified as normal as per the KNN algorithm, the data point (57, 170) should be normal.

Weight(x2)	Height(y2)	Class	Euclidean Distance
51	167	Underweight	6.7
62	182	Normal	13
69	176	Normal	13.4
64	173	Normal	7.6
65	172	Normal	8.2
56	174	Underweight	4.1
58	169	Normal	1.4
57	173	Normal	3
55	170	Normal	2

k = 3

date. 20/11/2024

Example-2

NAME	AGE	GENDER	CLASS OF SPORTS
Ajay	32	0	Football
Mark	40	0	Neither
Sara	16	1	Cricket
Zaira	34	1	Cricket
Sachin	55	0	Neither
Rahul	40	0	Cricket
Pooja	20	1	Neither
Smith	15	0	Cricket
Laxmi	55	1	Football
Michael	15	0	Football

Lily age 5, find out her class when k = 3

NAME	AGE	GENDER	CLASS OF SPORTS
Ajay	32	0	Football
Mark	40	0	Neither
Sara	16	1	Cricket
Zaira	34	1	Cricket
Sachin	55	0	Neither
Rahul	40	0	Cricket
Pooja	20	1	Neither
Smith	15	0	Cricket
Laxmi	55	1	Football
Michael	15	0	Football

Here male is denoted by 0 and female is 1. Lily's age 5 and gender female. (note. K = 3)

lily (x_2, y_2): lily (5, 1)

* rest of the data-set subtract from the lily for finding distance.

So we have to find out the distance using

$d=\sqrt{(x2-x1)^2+(y2-y1)^2}$ to find the distance between any two points.

So let's find out the distance between Ajay and Lily using formula

$d=\sqrt{(age2-age1)^2+(gender2-gender1)^2}$

$d=\sqrt{(5-32)^2+(1-0)^2}$

$d=\sqrt{729+1}$

$d=27.02$

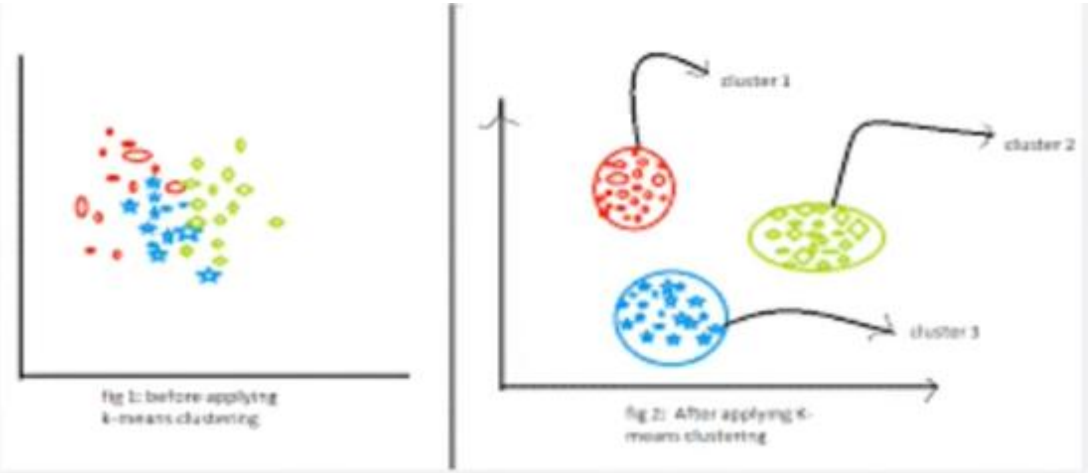
Distance between Angelina and	Distance
Ajay	27.02
Mark	35.01
Sara	11.00
Zaira	9.00
Sachin	50.01
Rahul	35.01
Pooja	15.00
Smith	10.00
Laxmi	50.00
Michael	10.05

Zaira	9	cricket
Michael	10	cricket
smith	10.5	football

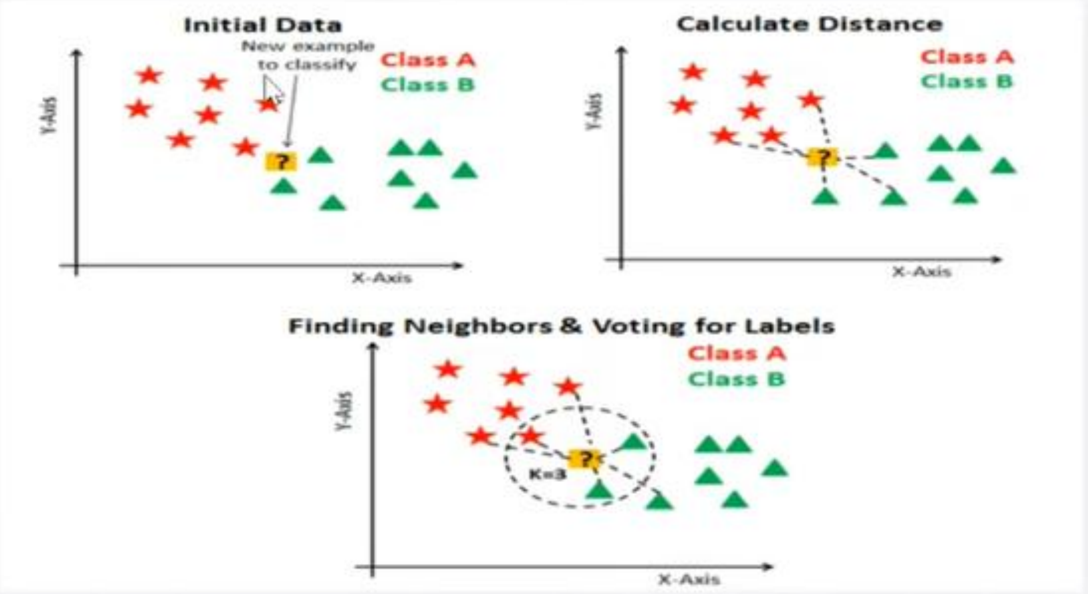
k = 3, so we will sort according to the least distance and then take the voting power for 3.

Difference between K-means and KNN algorithm

What is K-means clustering?
K- means clustering is used for analyzing and grouping data which does not include pre-labeled class or even a class attribute at all.



What is K-nearest neighbors?
K-nearest neighbors (KNN) is a type of supervised learning algorithm used for both regression and classification.



Choosing reason of K-means and KNN

K-Means work with unlabeled data

So---it is unsupervised algorithm

Weight(x2)	Height(y2)
51	167
62	182
69	176
64	173
65	172
56	174
58	169
57	173
55	170

Cluster 1	Cluster 2
Underweight(values)	Overweight(values)

KNN work with labeled data

So---it is supervised algorithm

Weight(x2)	Height(y2)	Class
51	167	Underweight
62	182	Normal
69	176	Normal
64	173	Normal
65	172	Normal
56	174	Underweight
58	169	Normal
57	173	Normal
55	170	Normal

57 kg	170 cm	?

K MEANS	K NN
Machine learning algorithm	Machine learning algorithm
Falls under unsupervised learning	Falls under supervised learning
Used for clustering	Used for classification, and regression
"K" in Kmeans represents number of class	"K" in KNN represents number of nearest neighbor
it creates classes	Classes are already created

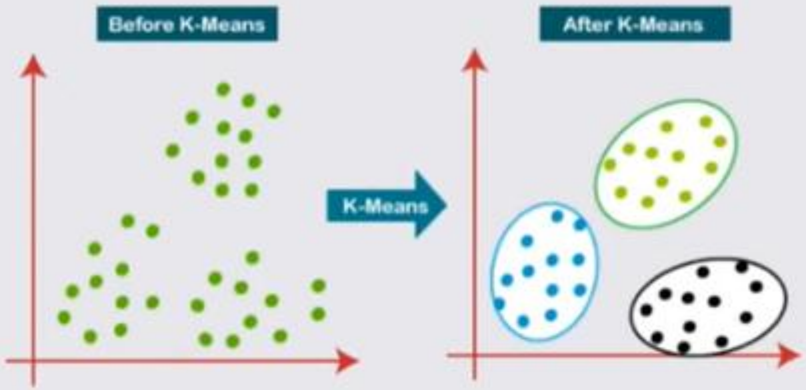
K-Means

K stands for how many cluster should be made.

Ex: 3-Means clustering means there should be 3 cluster which can perfectly group or cluster the whole dataset

Here the optimal value of K is find out by using Elbow method.

Here the value of K can be even .

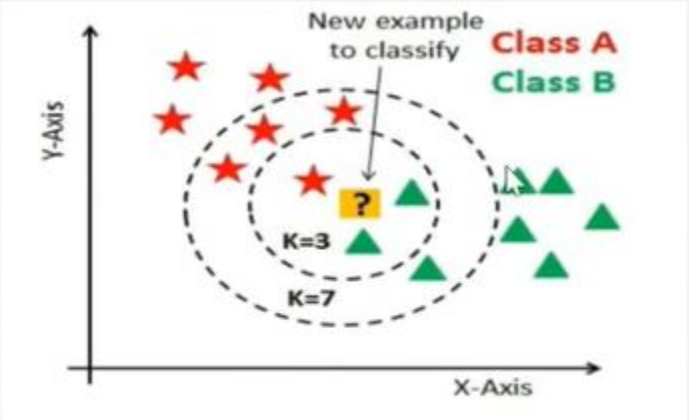


KNN

K stands for determine how many nearest data point or neighbors will vote for classification.

Ex: 3 –NN means there will be 3 nearest neighbor will vote for classification for new datapoint.

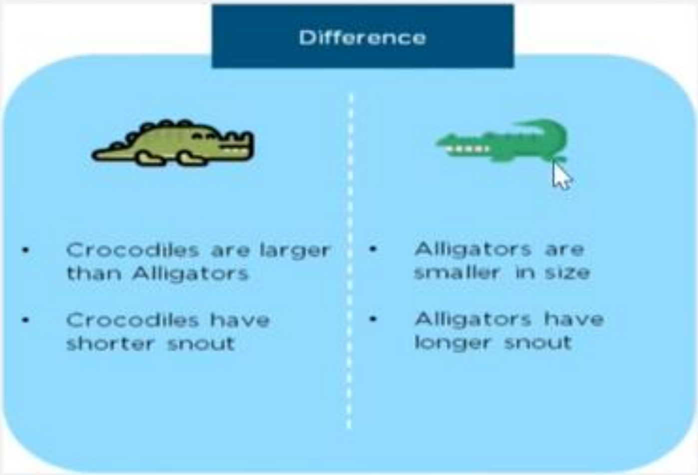
There is no particular way to determine the best value for "K",
But. .... Sqrt(n), where n is a total number of data points(if in case n is even we have to make the value odd by adding 1 or subtracting 1 that helps in select better)



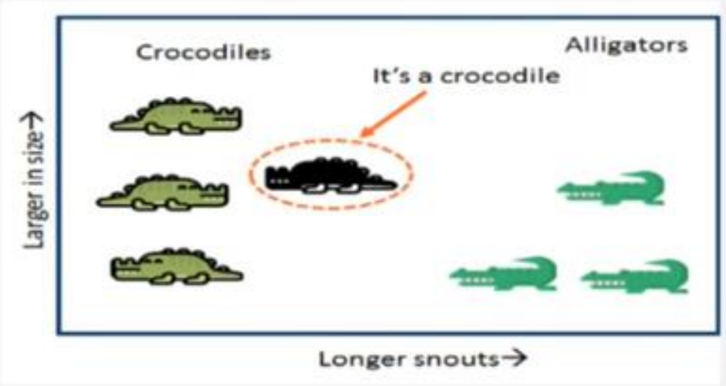
[redacted]

K-means and KNN ?

You can differentiate between a crocodile and an alligator based on their characteristics.



The features of the unknown animal are more a crocodile.
Therefore, it is a crocodile!

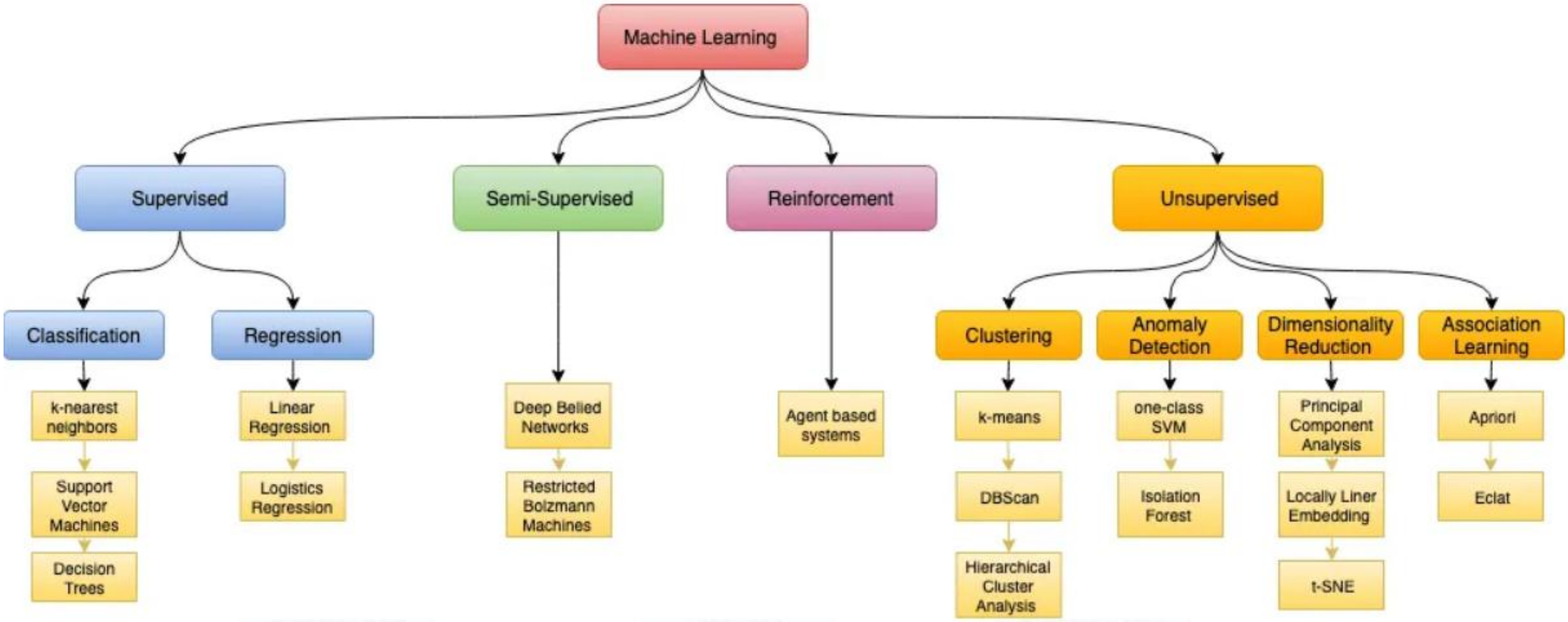


K-means and KNN ?

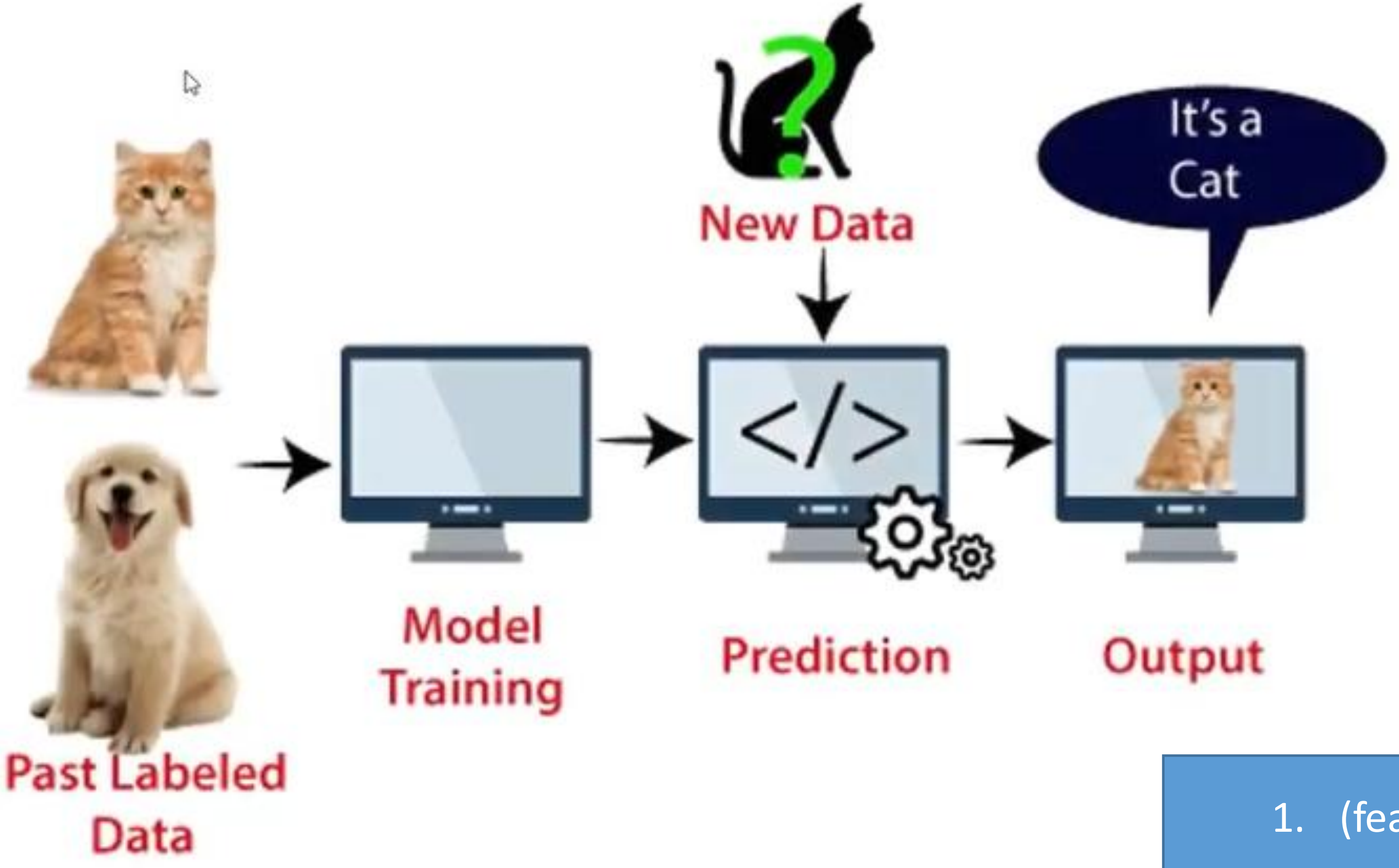
K-means and KNN ?

?

find out fruit name


Machine Learning
Supervised
Semi-Supervised
Reinforcement
Unsupervised
Classification
Regression
k-nearest neighbors
Support Vector Machines
Decision Trees
Linear Regression
Logistics Regression
Deep Belied Networks
Restricted Bolzmann Machines
Agent based systems
Clustering
Anomaly Detection
Dimensionality Reduction
Association Learning
k-means
DBScan
Hierarchical Cluster Analysis
one-class SVM
Isolation Forest
Principal Component Analysis
Locally Liner Embedding
t-SNE
Apriori
Eclat

Support Vector Machine Algorithm



1. (feature data) Vectori-zed
2. hyperplane

Uses:

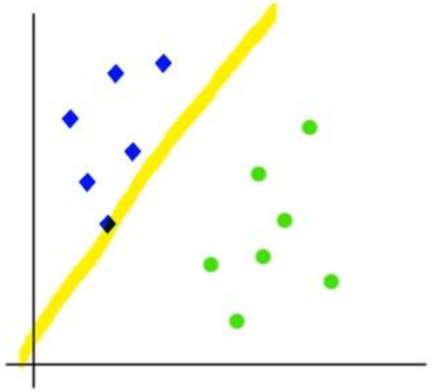
Face detection, image classification, text categorization

Types of SVM

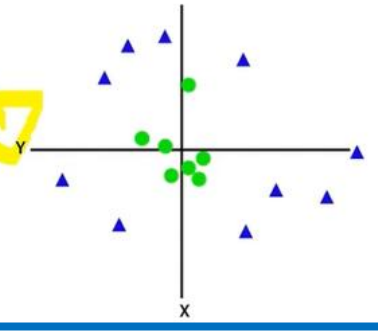
- TWO TYPES
  - **Linear SVM**
  - **Non-linear SVM**

Types of SVM

- TWO TYPES



- **Linear SVM**
- **Non-linear SVM**



Hyperplane and Support Vectors in the SVM algorithm:

- **Hyperplane:** Boundary is known as the hyperplane of SVM

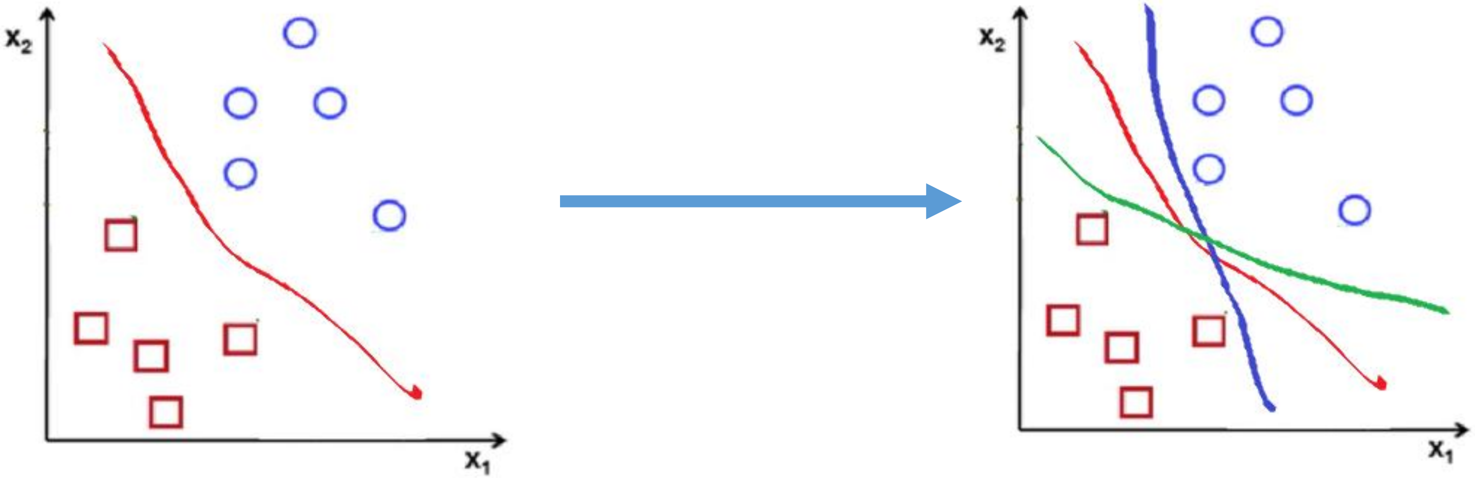
- **Support Vectors:**

Vectors support the hyperplane, hence called a Support vector.

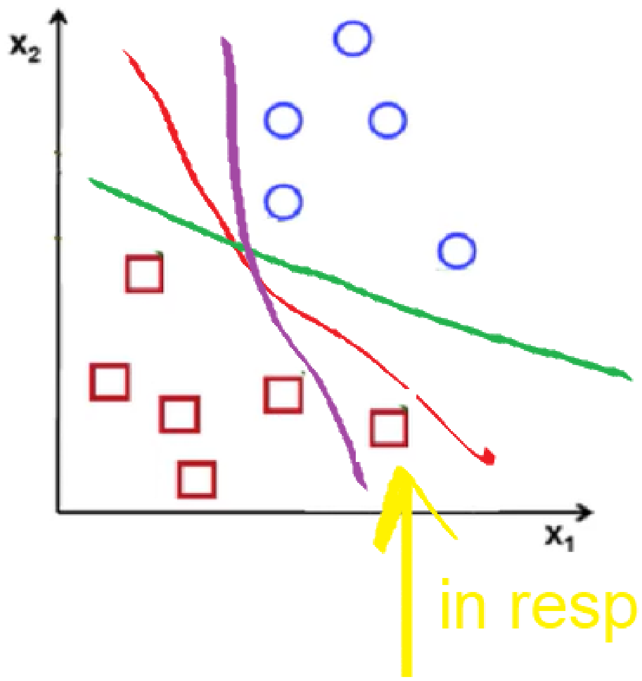
Support Vector Machine (SVM)

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

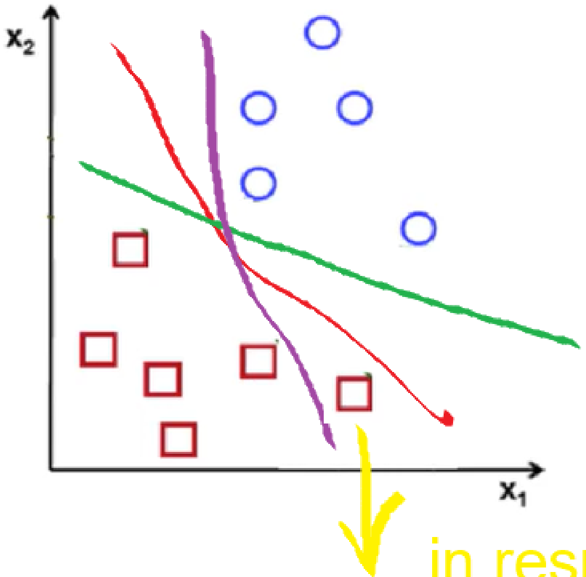


1



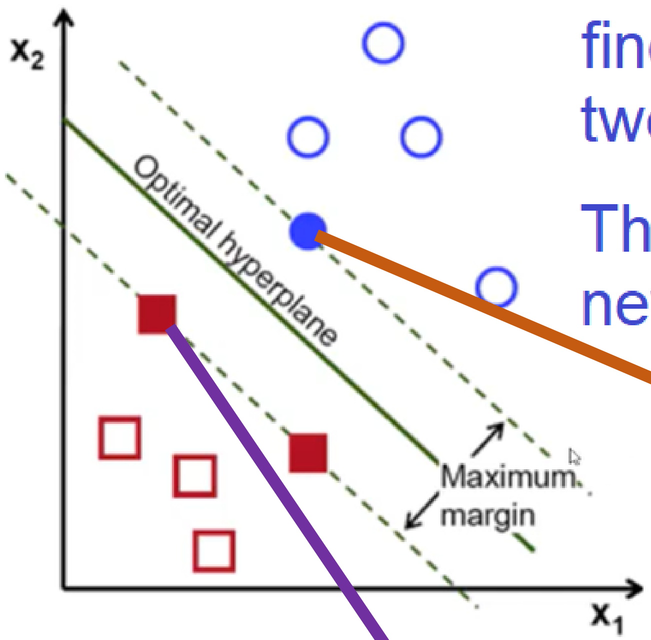
in respect of red line

2



in respect of purple line

S-1

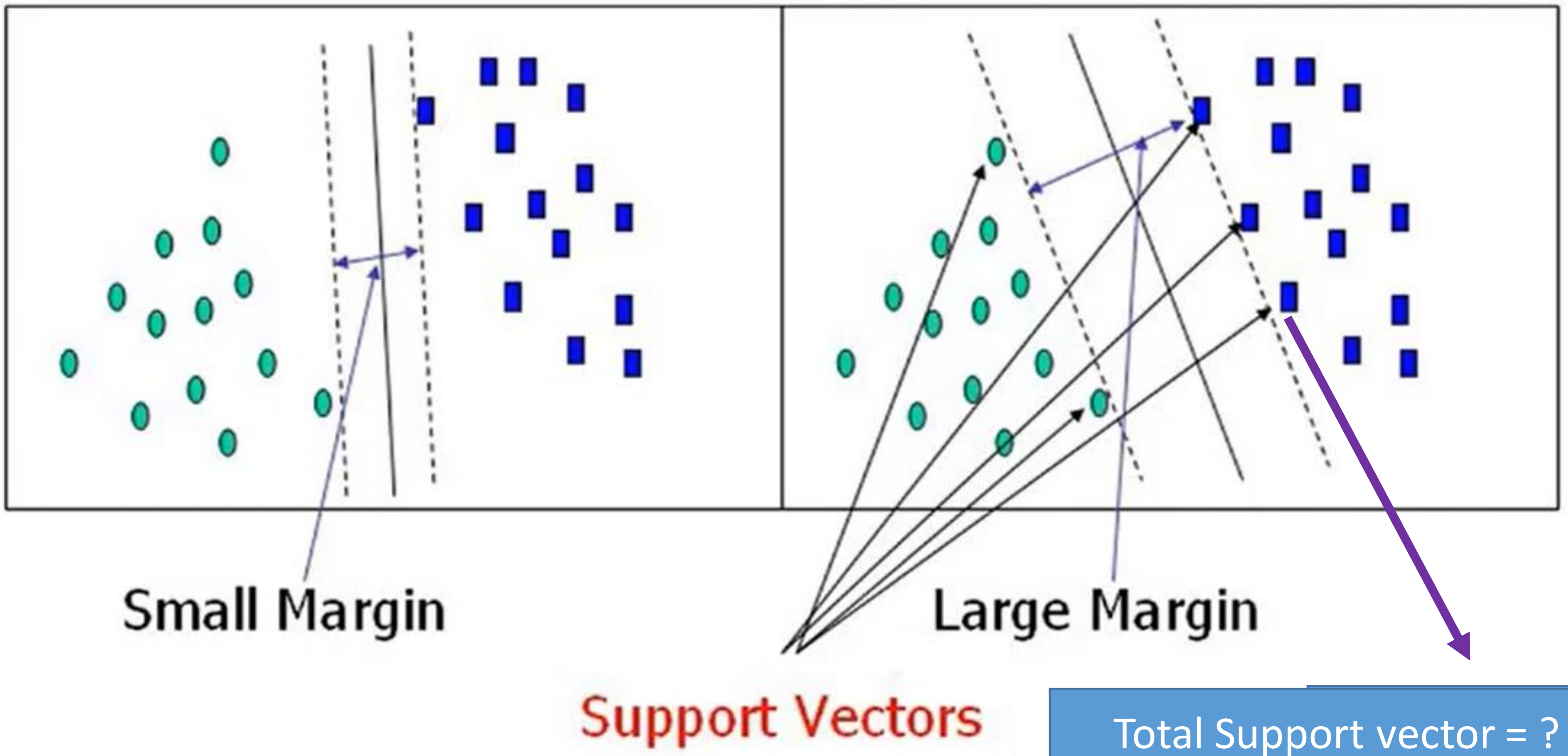


find out nearest distance between two classes, and take those points.

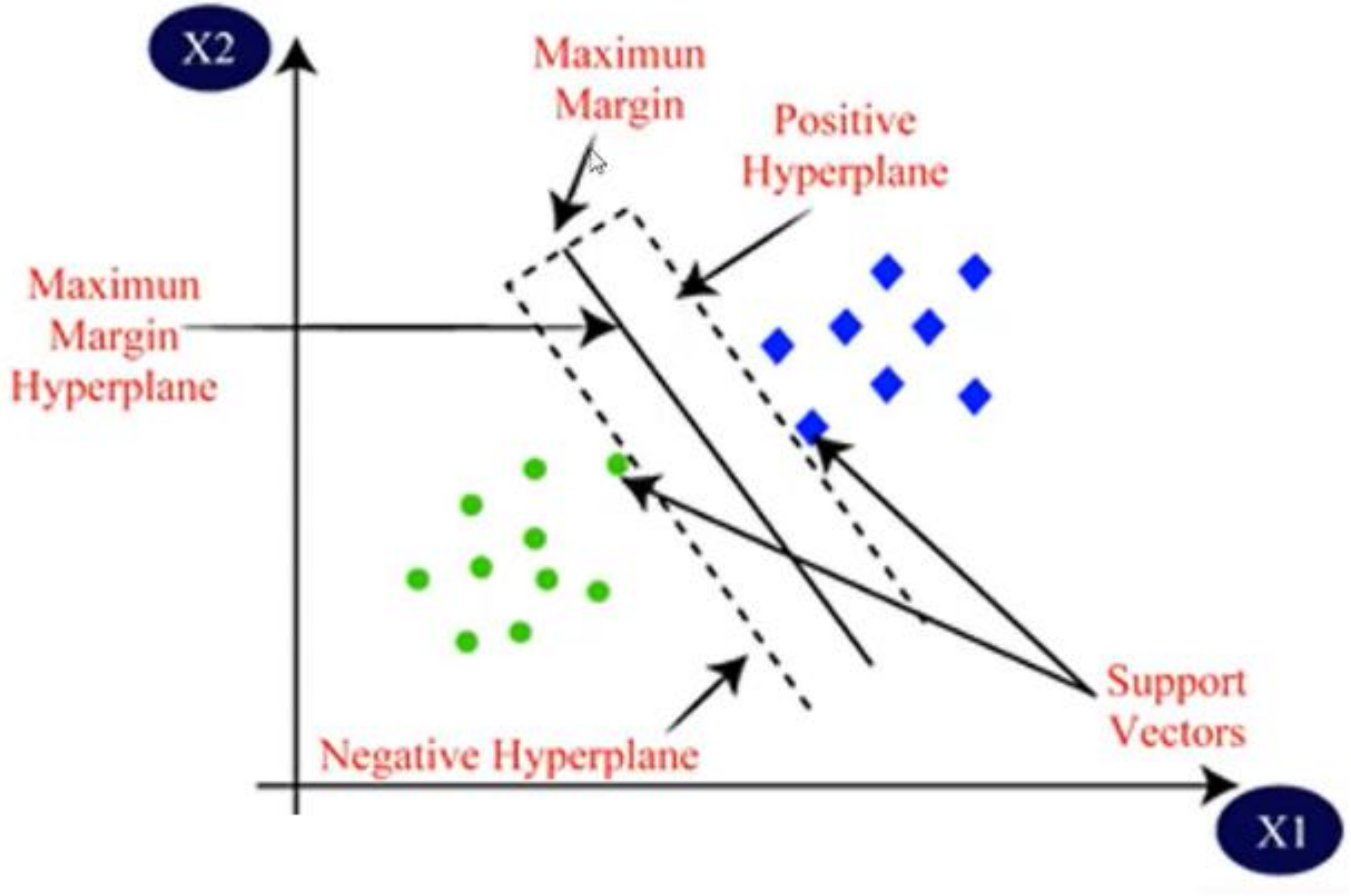
Then draw two lines and then draw a new line between those two lines.

Support vector

Support vector



Total Support vector = ?


X2
Maximun
Margin
Positive
Hyperplane
Maximun
Margin
Hyperplane
Support
Vectors
Negative Hyperplane
X1